

Lecture 13

Mantis shrimp vision: hyperspectral and polarization sensing

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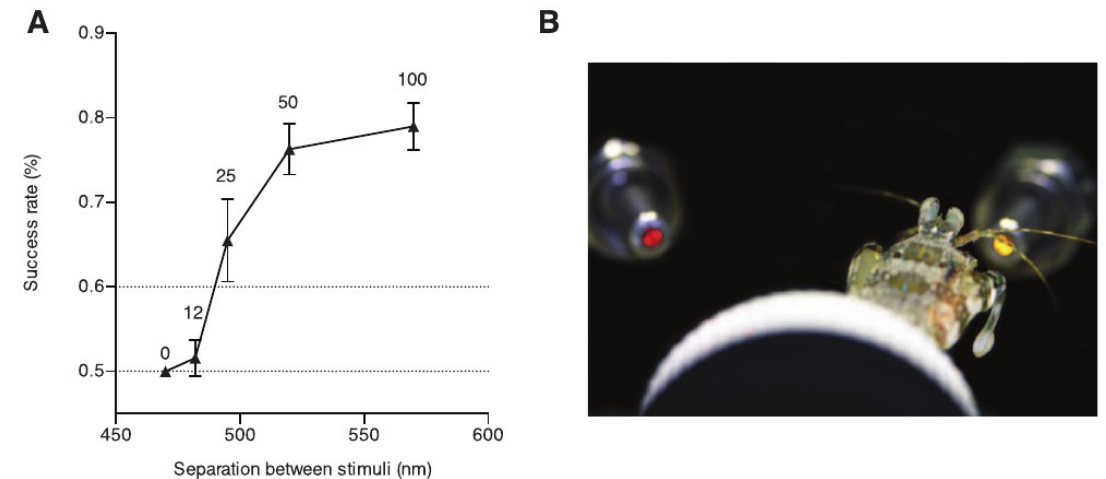
Bok et al. (2018), Fig. 1(a). Haptosquilla trispinosa in its natural burrow; original photograph by Roy Caldwell.

Learned choices measure wavelength discrimination

Mantis shrimps are stomatopods, marine crustacean predators with compound eyes. **Spectral discrimination** is the ability to distinguish lights of different wavelengths, measured in nanometers (nm). A retina with many narrow spectral channels is sometimes described as hyperspectral; the channel count alone does not specify the animal's discrimination threshold.

Thoen and colleagues trained *Haptosquilla trispinosa* to associate a particular wavelength with food. Two optical fibers presented narrow-band lights, and grabbing or tapping a fiber registered a choice. Test wavelengths varied relative to the rewarded wavelength.

Widely separated wavelengths supported successful choices. The assay measures a learned visual decision in a controlled feeding context, rather than the number of colors experienced subjectively or performance in every natural hunting task.



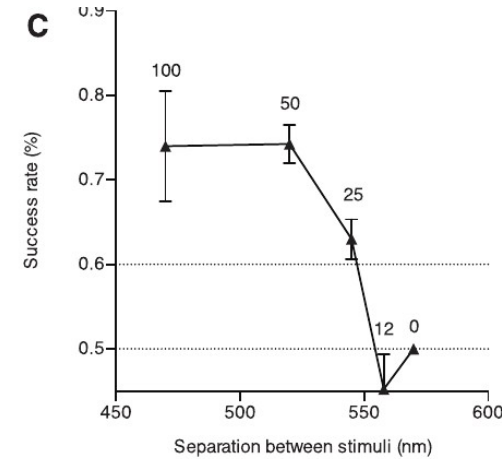
Thoen et al. (2014), Fig. 2(A, B). Food-associated wavelength choices in *Haptosquilla trispinosa*.

Rewarded wavelengths can be tested from both sides

A **narrow-band stimulus** concentrates its light within a limited wavelength range. Thoen and colleagues trained separate animals to lights between 400 and 650 nm, including 470 and 570 nm, then paired the trained light with an alternative delivered through another optical fiber.

Tests approached the trained wavelength from longer or shorter wavelengths. At separations of approximately 50–100 nm, correct choices were commonly 70–80%. As the alternatives approached the rewarded wavelength, performance declined toward the 50% chance level of a two-choice task.

Testing multiple trained wavelengths prevents one exceptionally easy color pair from defining the whole visual system. The response remains a food-associated choice; it does not reveal which receptor channel, interneuron or synapse supplied the decision signal.



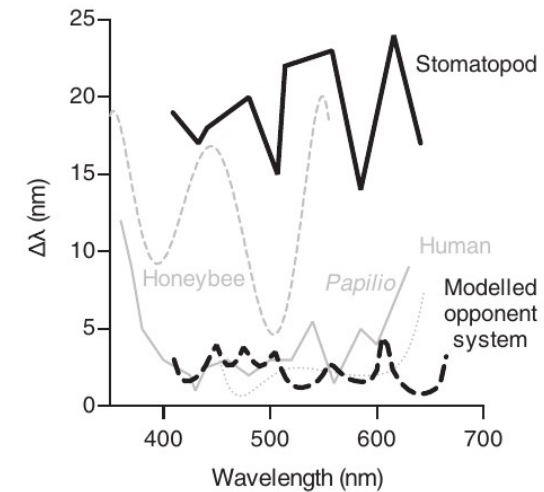
Thoen et al. (2014), Fig. 2(C, D). Choices toward shorter test wavelengths after 570-nm training.

Fine receptor tuning exceeds behavioral resolution

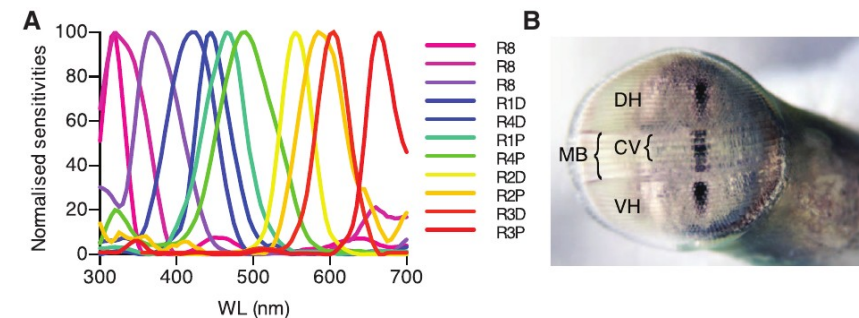
The **discrimination threshold** is the stimulus separation associated with a specified performance criterion. Thoen and colleagues used 60% correct choices to estimate the minimum discriminable wavelength difference in *Haptosquilla trispinosa*.

Behavioral thresholds fell approximately between 12 and 25 nm across the tested visible spectrum. Their receptor-noise model, which assumed comparisons between adjacent spectral channels, predicted substantially finer differences of approximately 1–5 nm. Intracellular recordings supplied the receptor sensitivities used in that prediction.

The mismatch challenges that specific efficient opponent-model account. It does not establish that every spectral comparison is absent: later behavioral evidence supports some opponent processing. Receptor tuning, model assumptions and measured discrimination are separate levels of evidence.



Thoen et al. (2014), Fig. 3. Measured wavelength thresholds and the authors' opponent-model prediction.



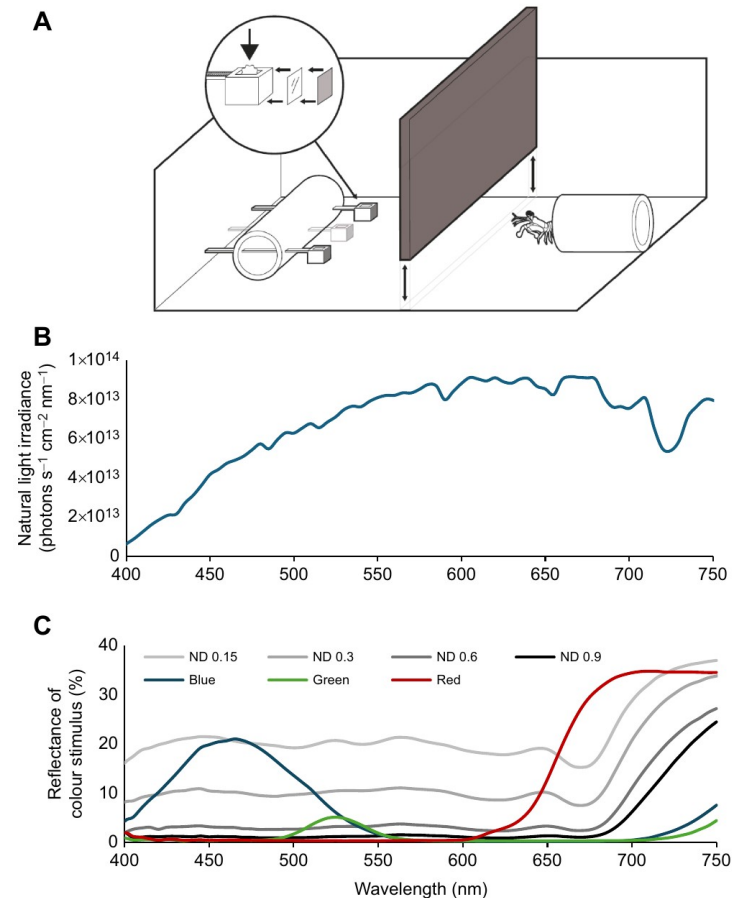
Thoen et al. (2014), Fig. 1(A, B). Recorded receptor sensitivities normalized to their peaks.

Grey distractors control brightness and target position

Color vision distinguishes spectral composition despite variation in overall intensity. Wang and Marshall tested *Haptosquilla trispinosa* with one colored cable-tie target and two grey distractors, using four neutral-density filter levels to vary distractor brightness.

Targets were about 4×4 mm and initially 5–8 cm from the animal. Positions and grey combinations varied across trials. Testing used new, uncontaminated cable ties; food was delivered only after a correct choice, reducing odor cues associated with the rewarded target.

Animals could approach and inspect the stimuli before grasping or tapping one. This design separates a spectral choice from a fixed side or simple brightness preference. It does not isolate the eyes from all movements or record the neurons that compare spectral inputs.



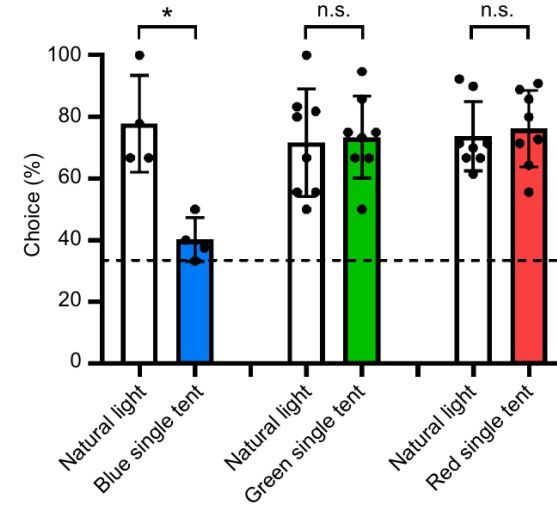
Wang & Marshall (2025), Fig. 3(A–C). Three-choice chamber, illumination and stimulus reflectance spectra.

Restricted illumination preserves some color choices

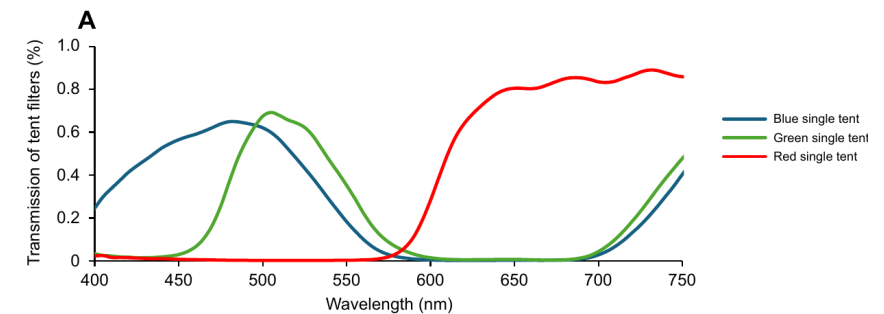
Spectral opponency compares signals from receptors with different wavelength sensitivities. Wang and Marshall restricted illumination with blue, green or red filter tents while trained *Haptosquilla trispinosa* chose colored targets against grey alternatives. Each group encountered its corresponding illumination color.

Under green and red illumination, correct choices remained about three-quarters of responses, above the one-third chance level of three alternatives. Blue-target performance fell from 76% under natural light to 40% under blue illumination. The spectral windows therefore produced different behavioral outcomes.

The authors interpret preserved green and red discrimination as evidence for opponent processing when only a restricted receptor subset receives substantial light. This supports spectral comparison, but neither identifies its neurons nor establishes the proposed hybrid system's complete mechanism.



Wang & Marshall (2025), Fig. 7. Color-versus-grey choices under natural and colored illumination.



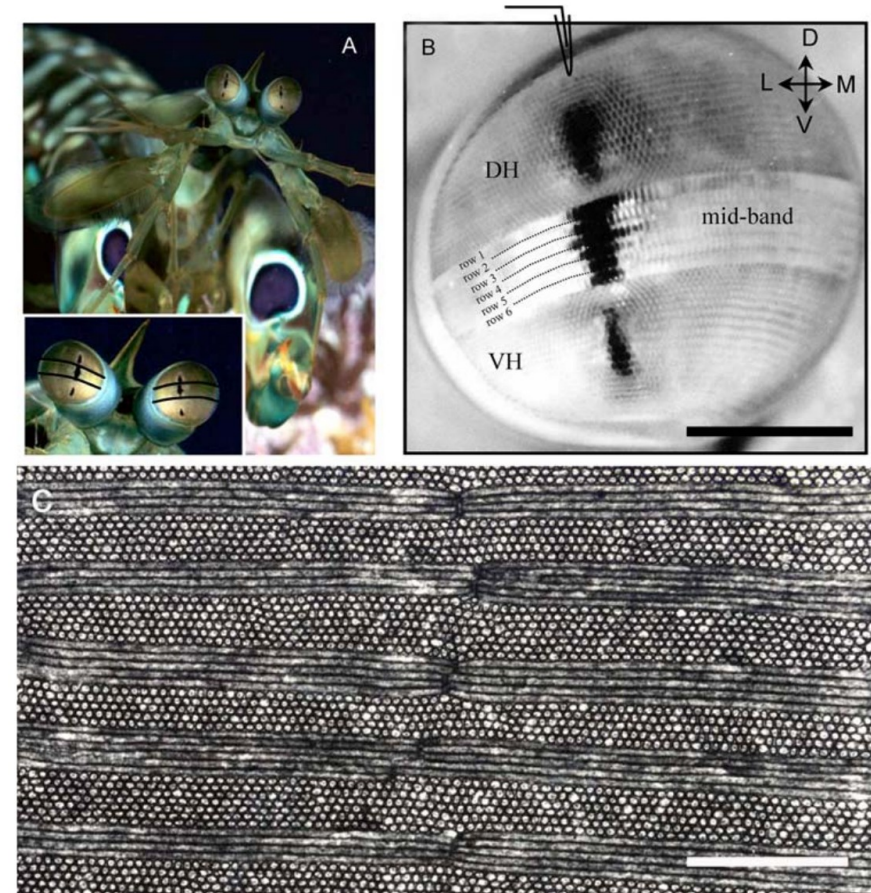
Wang & Marshall (2025), Fig. 5(A). Measured transmission of the illumination filters.

Three eye regions sample overlapping visual space

A **compound eye** contains repeated optical units called ommatidia. In six-rowed stomatopod eyes, a specialized midband separates the dorsal and ventral hemispheres. Each ommatidium contains photoreceptor cells whose light-sensitive membrane contributes to a central **rhabdom**.

Kleinlogel and White examined *Gonodactylus smithii* eyes and recorded from identified photoreceptors. The three dark **pseudopupils** arise where ommatidia facing the observer absorb light strongly. Their simultaneous appearance reflects overlap between the visual fields of the two hemispheres and the midband.

Overlap permits different receptor populations to sample related directions while supplying distinct spectral or polarization signals. The optical arrangement is anatomical evidence; it does not establish that every downstream neuron combines all three regions or that all stomatopod species share identical specialization.



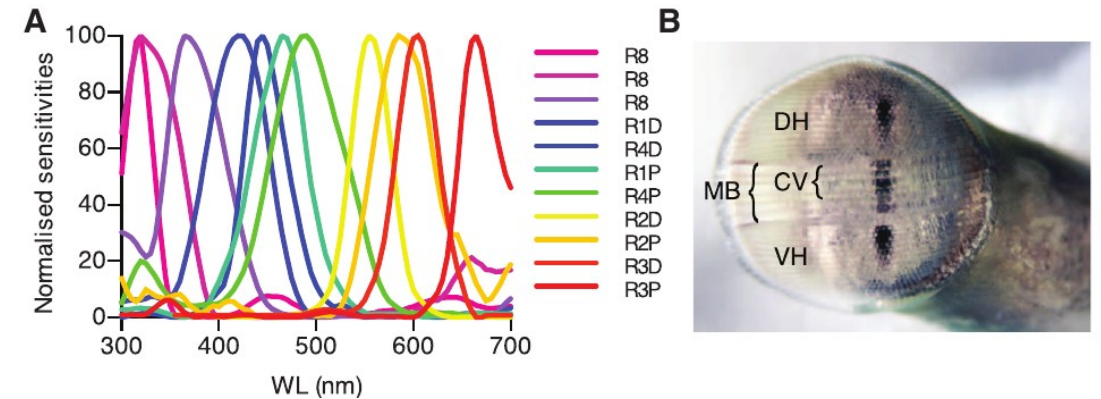
Kleinlogel & White (2008), Fig. 3(A–C). The animal, eye pseudopupils and midband ultrastructure.

Midband rows separate spectral and polarization inputs

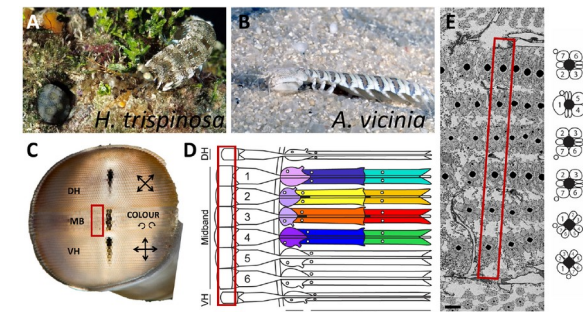
The **midband** is the specialized strip of enlarged ommatidia separating the eye's hemispheres. In *Haptosquilla trispinosa*, its first four rows carry differently tuned spectral receptors; the last two rows participate in circular or elliptical polarization sensing.

Thoen and colleagues recorded eight visible-sensitive receptor classes and three ultraviolet-sensitive classes; another ultraviolet class was difficult to locate. Streets and colleagues mapped the six-row retinal organization with serial electron microscopy and reconstructed the associated photoreceptor projections.

Anatomical row identity and recorded sensitivity are complementary measurements. Twelve potential spectral channels in the color rows do not imply twelve independently established perceptual dimensions. Functional assignments also require species-specific evidence rather than assuming that every mantis shrimp has the same six-row system.



Thoen et al. (2014), Fig. 1(A, B). *Haptosquilla* eye with dorsal, ventral and color-row labels.



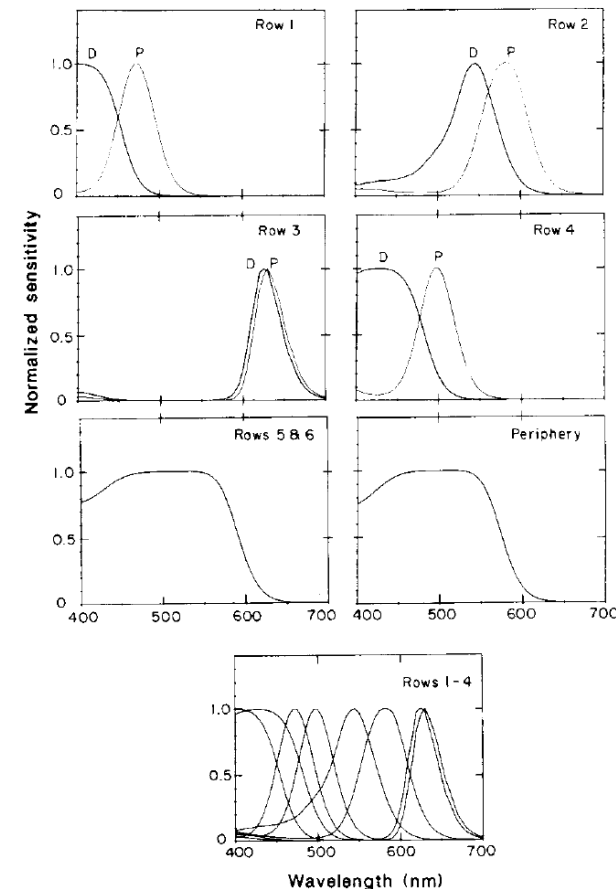
Streets et al. (2025), Fig. 1(A–E). Published retinal layout and electron-microscopy cross-section.

Three optical tiers place receptors at different depths

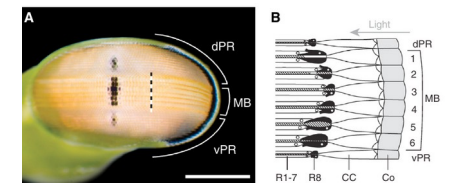
A **retinular cell** is a photoreceptor forming part of an ommatidium. In color midband rows, the distal R8 receptor lies nearest the incoming light, while cells R1–R7 contribute to two deeper tiers. **Distal** means nearer the optical entrance; **proximal** means farther along the light path.

Cronin and Marshall measured pigments and filters in frozen *Pseudosquilla ciliata* retinal sections. Each of the eight main-rhabdom tiers in rows 1–4 contained a distinct pigment. Light entering a deeper tier had already passed through the overlying material.

Tiering therefore changes both receptor position and the available spectrum. The published sensitivity curves combine measured absorption with optical assumptions; they are not direct recordings of every receptor's voltage response.



Cronin & Marshall (1989), Fig. 3. Predicted sensitivities of distal and proximal retinal tiers.



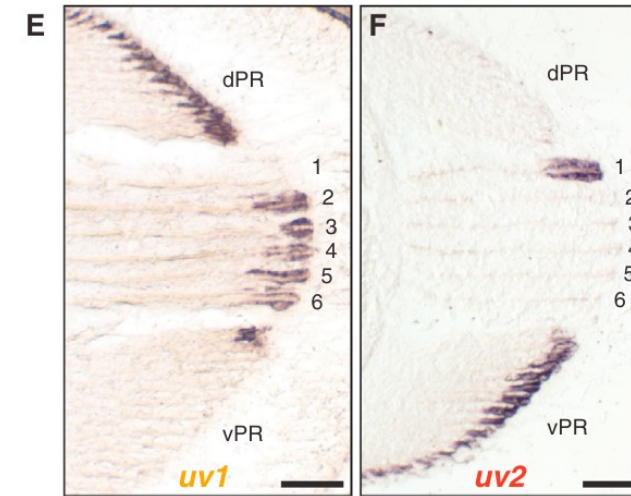
Bok et al. (2014), Fig. 1(A, B). Eye surface and original sagittal retinal diagram.

R8 combines ultraviolet reception with optical filtering

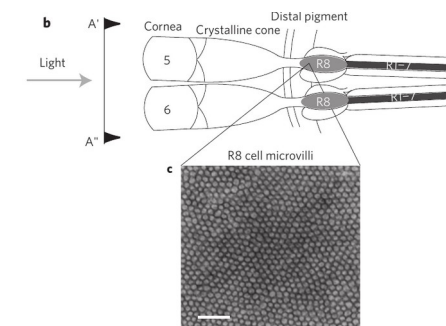
The **R8 cell** is the eighth retinular photoreceptor, located above the main R1–R7 rhabdom. Its optical position allows it to absorb ultraviolet photons while altering visible light transmitted toward deeper receptors. These operations depend on the row and species.

In *Neogonodactylus oerstedii*, Bok and colleagues localized ultraviolet opsin transcripts to R8 cells. In *Odontodactylus scyllarus*, Roberts and colleagues examined unusually long, ordered R8 microvilli in polarization rows 5 and 6, where the rhabdomere was approximately 150 μm long.

A **rhabdomere** is one cell's contribution to the light-sensitive rhabdom. R8 ultraviolet sensitivity and visible-light retardance are different properties of that structure. Neither implies that an ultraviolet electrical response is required for every visible circular-polarization signal.



Bok et al. (2014), Fig. 1(E, F). UV1 and UV2 transcript localization in retinal R8 cells.



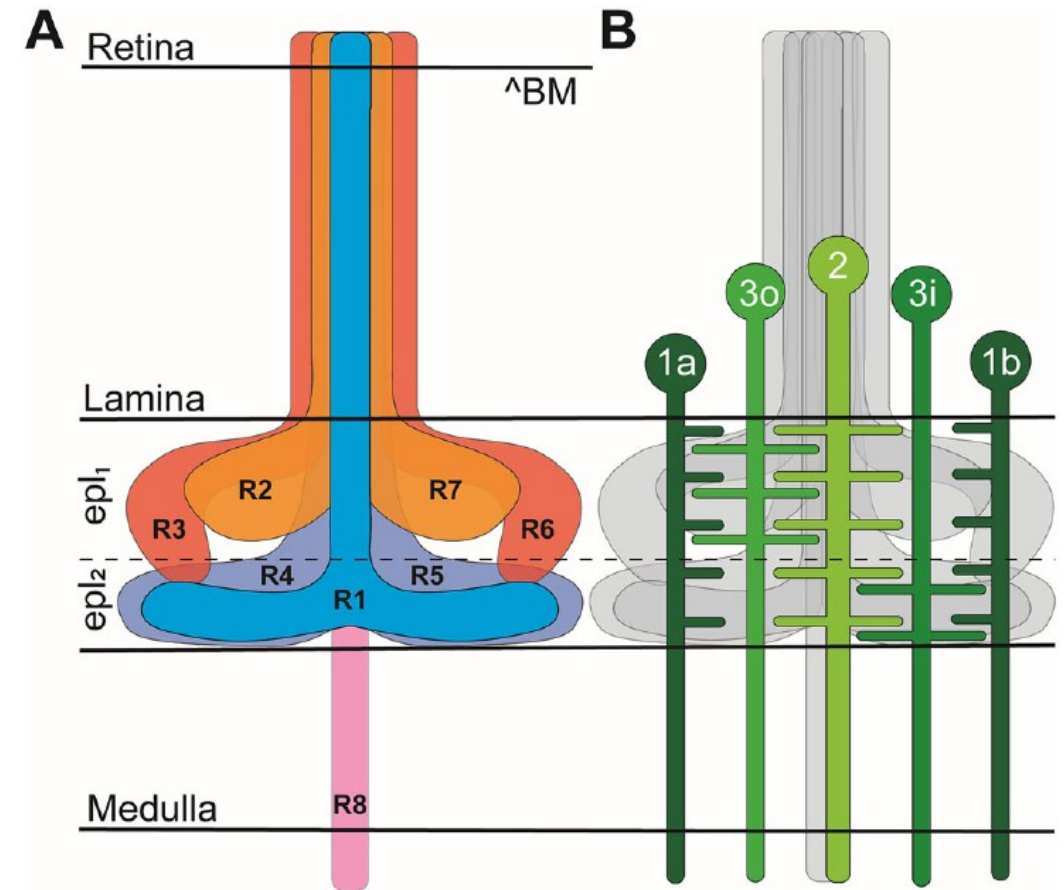
Roberts et al. (2009), Fig. 1(b, c). Rows 5–6 R8 position and microvillar ultrastructure.

Photoreceptor axons reach distinct lamina layers

The **lamina** is the first optic neuropil, a region containing neuronal processes where retinal signals encounter interneurons. A **lamina cartridge** is the processing column associated with an ommatidium. Streets and colleagues reconstructed these columns in *Haptosquilla trispinosa* and *Alachosquilla vicina*.

R2, R3, R6 and R7 terminals occupied the outer external plexiform layer, whereas R1, R4 and R5 occupied the inner layer. R8 axons extended through the lamina toward the **medulla**, the next optic neuropil. The reconstructions revised older assignments of the two receptor groups.

Layer separation preserves distinct input populations and supplies candidate sites for comparison. Terminal position alone does not establish synaptic direction, transmitter identity or an excitatory-versus-inhibitory response.



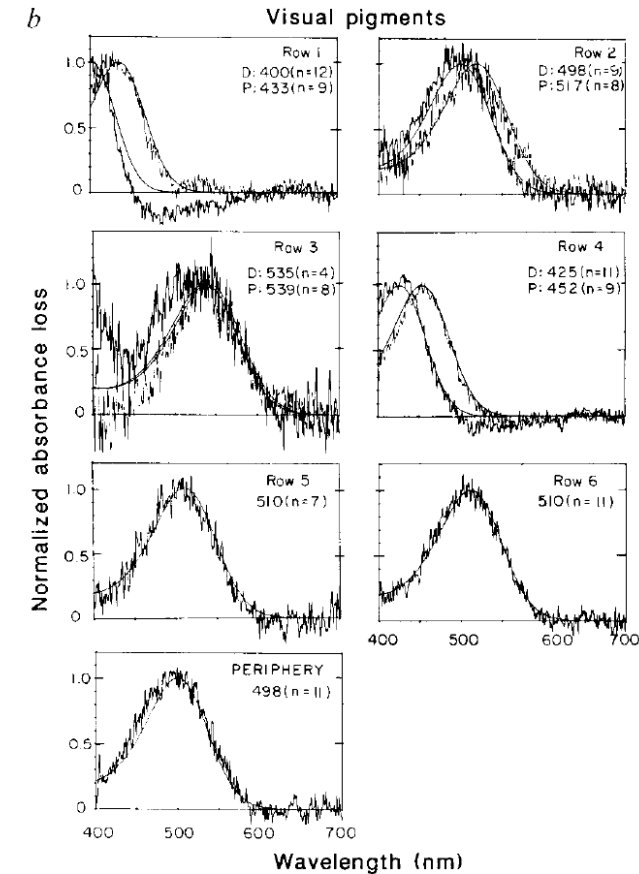
Streets et al. (2025), Fig. 2(A, B). Original reconstruction-based layout of receptor terminals and monopolar cells.

Microspectrophotometry separates pigment absorption

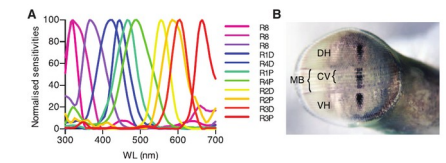
Microspectrophotometry measures how a microscopic specimen absorbs light across wavelengths. Cronin and Marshall directed a 1.5- μm measuring spot along the light-sensitive structures of frozen *Pseudosquilla ciliata* retinal sections, scanning from 400 to 700 nm.

Animals were dark-adapted before sampling. Absorption before and after strong illumination was compared in the same receptor region to isolate the light-sensitive pigment contribution. Fitted pigment peaks ranged from approximately 400 to 539 nm across the main retinal classes.

This procedure distinguishes pigment absorption from an intact cell's effective spectral sensitivity, which also depends on optical filtering and receptor length. The ultraviolet-sensitive R8 pigment was not characterized by this visible-range measurement, so its absence from these curves is not evidence that ultraviolet reception is absent.



Cronin & Marshall (1989), Fig. 2(b). Measured visual-pigment difference spectra with fitted templates.



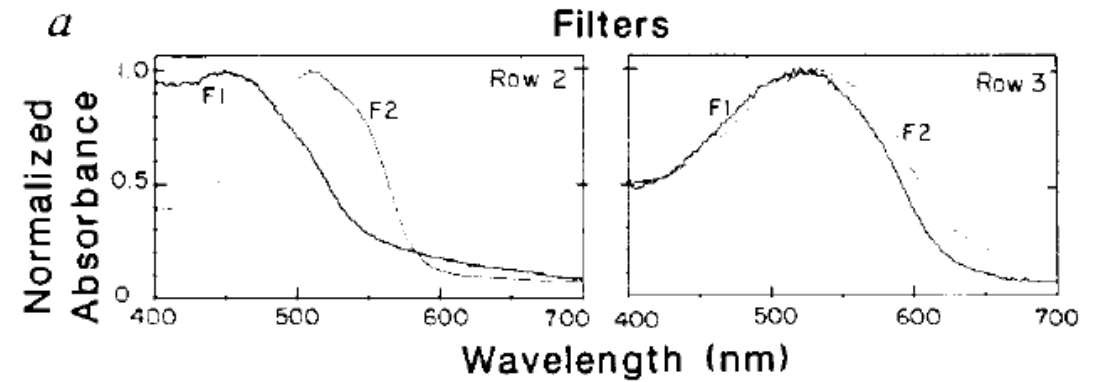
Thoen et al. (2014), Fig. 1(A, B). Later intracellular sensitivity curves in *Haptosquilla*.

Visual pigments and stable filters are distinct components

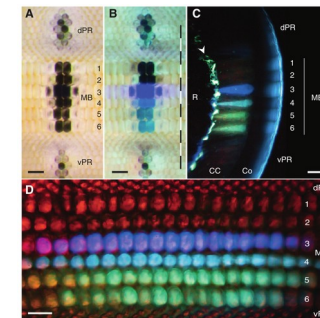
A **visual pigment** is a light-absorbing chromophore bound to an opsin protein; photon absorption initiates receptor activation. An **optical filter** selectively removes wavelengths from the light reaching that pigment. These components can therefore change sensitivity through different mechanisms.

Cronin and Marshall measured light-sensitive pigments and comparatively stable intrarhabdomal filters separately in *Pseudosquilla ciliata*. Midband rows 2 and 3 contained two filters in series. Bok and colleagues later identified a distinct ultraviolet-filter system in the crystalline cones of *Neogonodactylus oerstedii*.

Pigment diversity and filter diversity must not be treated as interchangeable counts. Filtering can produce different effective sensitivities without adding a new functional opsin, while separate visual pigments can also contribute distinct spectral channels.



Cronin & Marshall (1989), Fig. 2(a). Absorption spectra of the distal and proximal intrarhabdomal filters.



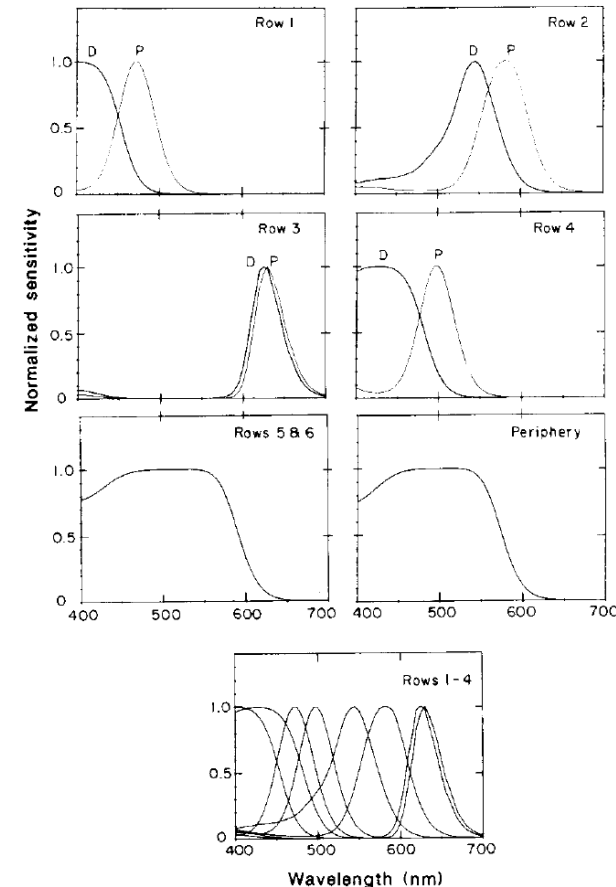
Bok et al. (2014), Fig. 2(A–D). Native UV fluorescence localizes a separate cone-filter system.

Successive filters narrow the spectrum reaching deep tiers

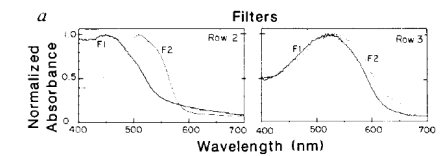
Serial filtering occurs when light passes through successive absorbing structures before reaching a receptor. Cronin and Marshall measured a distal filter and a proximal filter in each of midband rows 2 and 3 in *Pseudosquilla ciliata*.

The distal filter transmitted a broader wavelength range than the deeper filter. The underlying receptor tiers therefore received different spectra even within one optical unit. The authors calculated effective sensitivities by combining each tier's pigment absorption with transmission through the material above it.

The resulting predicted sensitivity functions were narrower than the pigment absorption curves alone. This optical mechanism supplies selective spectral inputs; it does not establish the downstream comparison rule or prove that behavior uses every distinction available at the receptor level.



Cronin & Marshall (1989), Fig. 3. Published calculations of sensitivity after serial filtering.



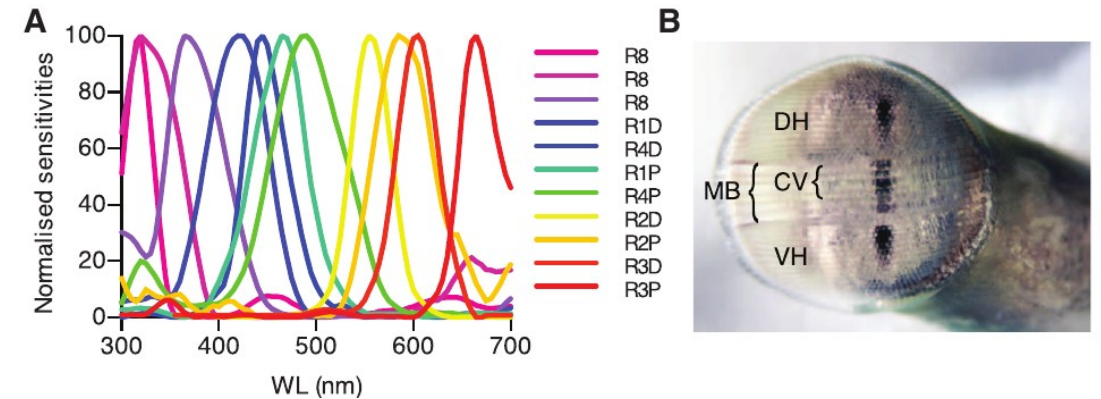
Cronin & Marshall (1989), Fig. 2(a). Distal and proximal filter spectra in rows 2 and 3.

Intracellular recordings measure effective spectral tuning

An **intracellular recording** measures the electrical potential inside a cell relative to the extracellular space. Thoen and colleagues used receptor recordings in *Haptosquilla trispinosa* to characterize the sensitivity functions supporting their discrimination analysis.

Eight measured classes peaked in the visible spectrum, and three more were recorded in the ultraviolet. The published curves are normalized to each class's maximum response, so their heights do not represent equal absolute responses to the same photon flux. An additional ultraviolet class was difficult to record.

These measurements include the consequences of the intact receptor's optical environment. They establish multiple effective spectral inputs, while leaving unresolved how channel signals are weighted, compared or recognized during a behavioral decision.



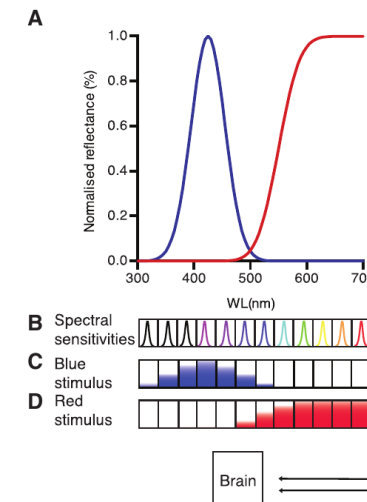
Thoen et al. (2014), Fig. 1(A, B). Receptor sensitivities measured by intracellular electrophysiology.

Serial color recognition remains a processing hypothesis

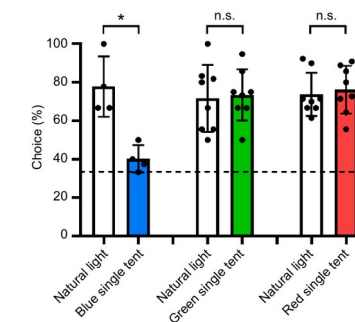
Color binning is the proposal that an activation pattern across discrete spectral channels identifies a color without fine analog comparisons between neighboring channels. Thoen and colleagues proposed that scanning the midband across an object could convert its spectrum into a temporal activation sequence.

Their approximately 12–25 nm behavioral thresholds were much coarser than predictions from a particular opponent model. The serial-recognition account was offered to explain that discrepancy; the study did not record a neuron decoding the proposed sequence.

Wang and Marshall's later color-versus-grey results support spectral comparison under restricted illumination. A hybrid of channel recognition and opponency is therefore plausible, but its architecture and temporal computations remain unresolved. Coarse discrimination does not justify excluding every opponent mechanism.



Thoen et al. (2014), Fig. 4(A–D). The authors' published proposal for spectral-channel recognition.



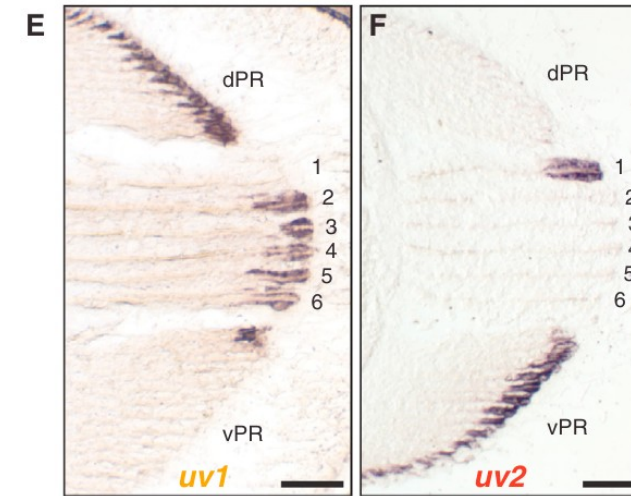
Wang & Marshall (2025), Fig. 7. Later color-versus-grey behavior under restricted illumination.

Two ultraviolet pigments support several receptor classes

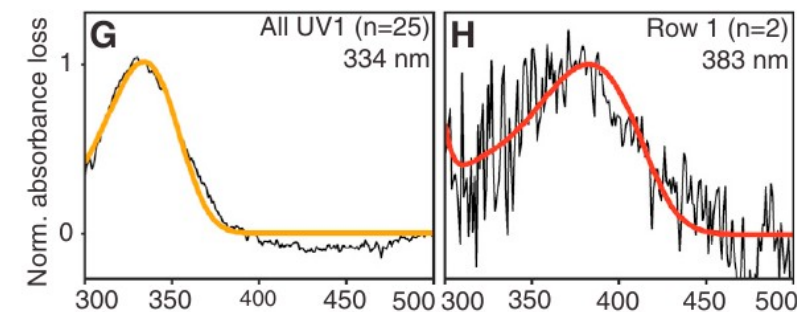
Ultraviolet light has wavelengths below approximately 400 nm. Bok and colleagues investigated the several ultraviolet receptor classes of *Neogonodactylus oerstedii* using retinal transcript sequencing, pigment measurements and **in situ hybridization**, which localizes particular RNA sequences in tissue.

The *uv1* transcript occupied R8 cells in midband rows 2–6, while *uv2* occupied row 1 and a complementary peripheral distribution. Microspectrophotometry measured corresponding pigment peaks near 334 and 383 nm. A third transcript did not establish a third functional pigment.

The two functional pigments were insufficient by themselves to explain the receptor-class diversity. Different filters acting on the same pigment supply additional tuning. Transcript detection, pigment absorption and electrical sensitivity are distinct measurements rather than equivalent evidence of a complete sensory channel.



Bok et al. (2014), Fig. 1(E, F). Separate UV1 and UV2 transcript distributions in R8.



Bok et al. (2014), Fig. 1(G, H). Measured pigment peaks at 334 and 383 nm.

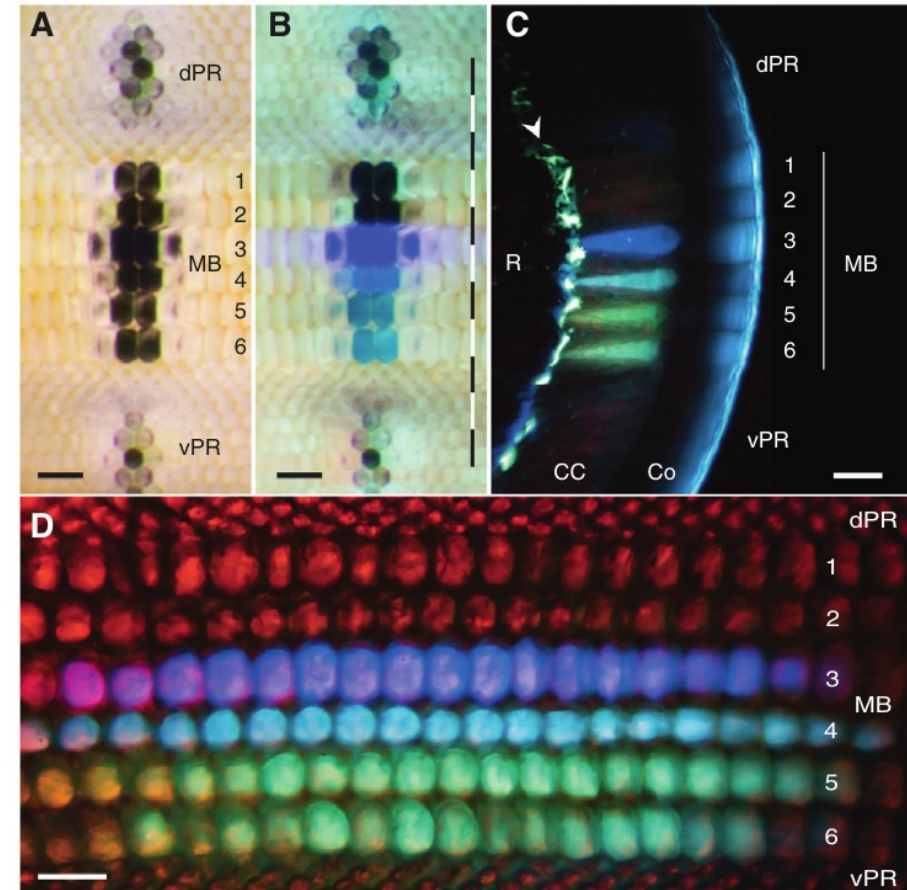
Ultraviolet fluorescence locates crystalline-cone filters

The **crystalline cone** is an optical structure between the cornea and the receptor rhabdom. Bok and colleagues illuminated *Neogonodactylus oerstedii* eyes with 375-nm light and observed native fluorescence concentrated in particular midband rows.

Fluorescence is emission following absorption of shorter-wavelength light.

Cross-sections localized the fluorescent material to the crystalline cones, directly in the path of light reaching R8. Microspectrophotometry then measured ultraviolet-absorbing pigments there. Peripheral cones and the cornea lacked comparable optically relevant ultraviolet absorption under the reported conditions.

Fluorescence provided a localization clue rather than a direct measure of receptor voltage. Absorption spectra and tissue position together support an optical-filter function; the visible fluorescent hue is not the wavelength that the underlying ultraviolet receptor necessarily detects.



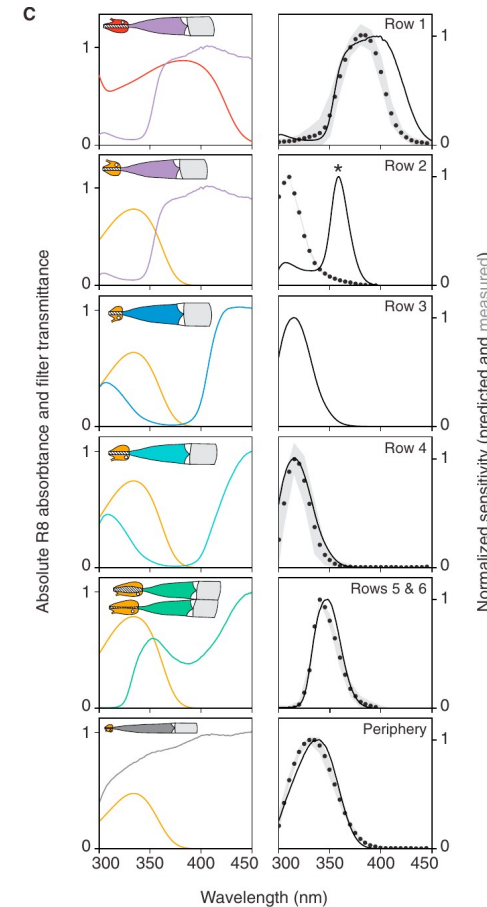
Bok et al. (2014), Fig. 2(A–D). White-light view, UV fluorescence and cone-layer cross-sections.

Cone filters shift ultraviolet sensitivity in both directions

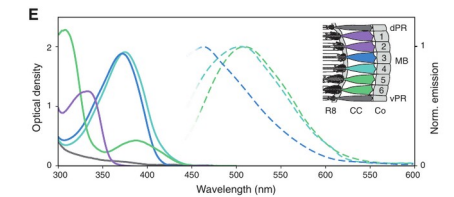
A **long-pass filter** removes shorter wavelengths while transmitting longer ones; a **short-pass filter** preferentially removes longer wavelengths. Bok and colleagues combined the measured cone-filter transmission with R8 pigment absorption in *Neogonodactylus oerstedii*.

Row 1 filtering reduced sensitivity below about 350 nm. Filters absorbing near 372 and 375 nm in rows 3 and 4 narrowed sensitivity toward shorter ultraviolet wavelengths. Two pigments in rows 5 and 6 produced a narrow transmitted band around a predicted receptor peak of 340 nm.

Most predictions agreed with earlier electrophysiology, but row 2 differed markedly. The authors proposed an earlier row-identification error because that recorded cell lacked histological confirmation. The discrepancy remains a qualification, not a measured reassignment of the original cell.



Bok et al. (2014), Fig. 3(C). Published predictions compared with measured receptor sensitivities.



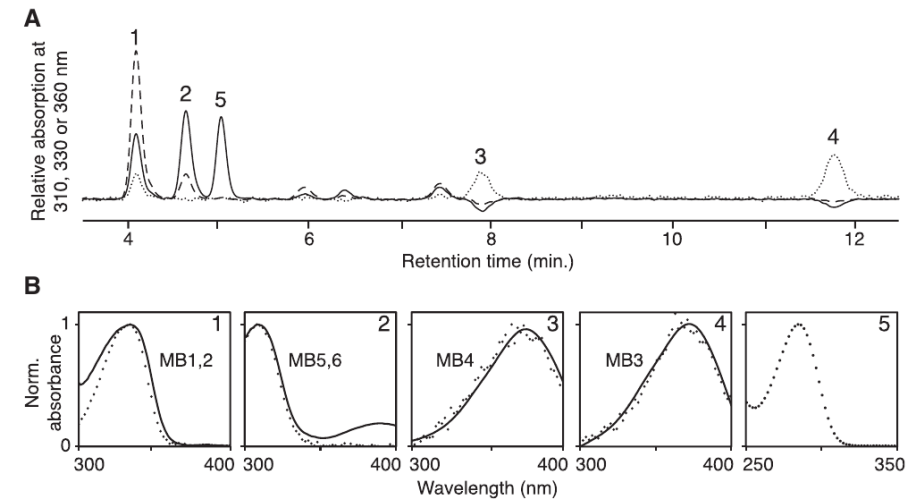
Bok et al. (2014), Fig. 2(E). Measured absorption and emission spectra of cone pigments.

Chemical analysis identifies sunscreen-derived filters

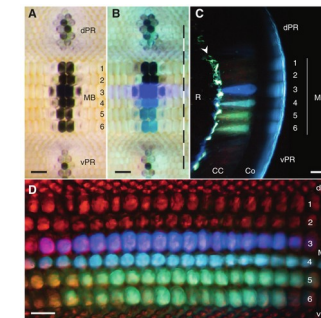
Mycosporine-like amino acids are small ultraviolet-absorbing compounds often associated with biological photoprotection. Bok and colleagues extracted cone pigments from *Neogonodactylus oerstedii* and used chromatography and mass spectrometry to separate them and examine their molecular properties.

The extracts contained several ultraviolet-absorbing peaks. Two matched the compounds porphyra-334 and mycosporine-gly, consistent with filter absorption near 334 and 308 nm. Other pigments resembled members of the same chemical family, but their complete structures were unresolved.

Molecules associated with sunscreen protection can thus contribute to spectral tuning when positioned before a receptor. The proposed dietary acquisition and subsequent modification of these compounds were not directly established. The study identifies optical and chemical components without explaining every step of pigment transport or biosynthesis.



Bok et al. (2014), Fig. 4(A, B). Chromatographic peaks and absorption spectra of extracted UV pigments.



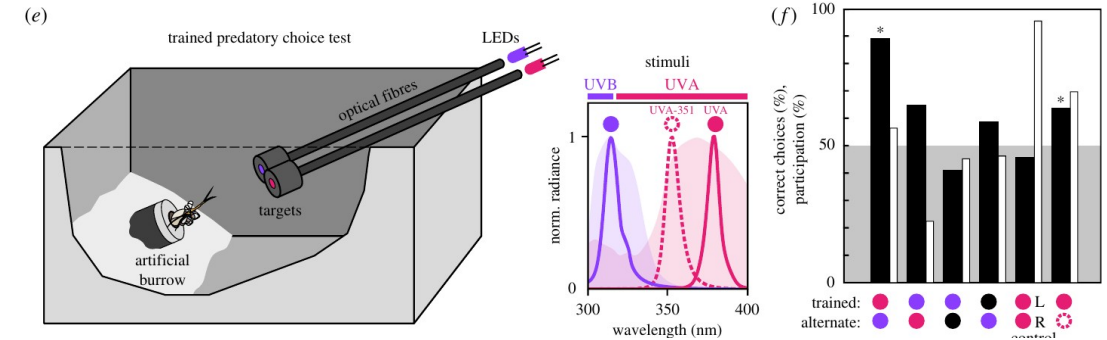
Bok et al. (2014), Fig. 2(A–D). The cone structures containing the measured pigments.

UV choices depend on wavelength beyond brightness

Bok and colleagues trained *Haptosquilla trispinosa* to choose ultraviolet targets delivered through optical fibers. LEDs peaked at 378.3 nm, 351.1 nm or 314.3 nm, spanning long- and short-wavelength portions of the ultraviolet range. Target positions varied, and relative brightness was changed across tests.

Animals trained to the 378.3-nm target selected it over the 314.3-nm target in about 90% of choices. They also distinguished 378.3 from 351.1 nm, a separation of 27.2 nm. Identical-target tests did not produce a consistent side preference.

Wavelength-dependent choices persisted despite the brightness manipulations, supporting ultraviolet spectral discrimination. Training to the short-wavelength ultraviolet target was much less successful, so the results do not imply symmetric learning or equal behavioral value across every ultraviolet channel.



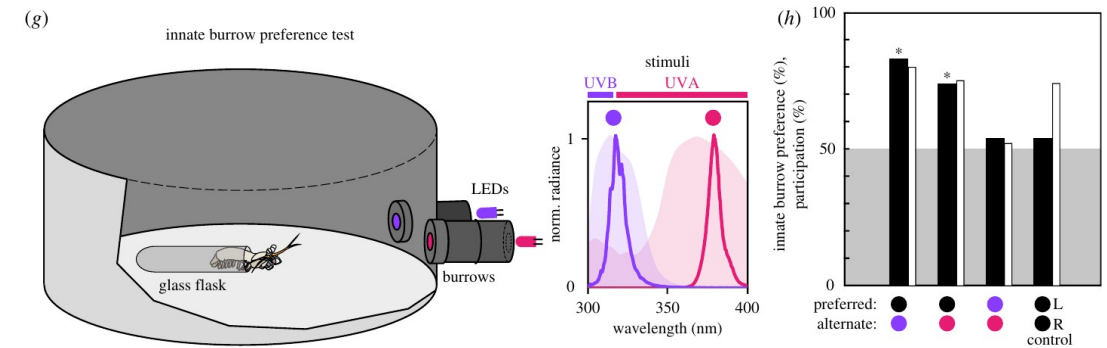
Bok et al. (2018), Fig. 1(e, f). Original trained-choice apparatus, LED spectra and responses.

Burrow choice separates detection from approach

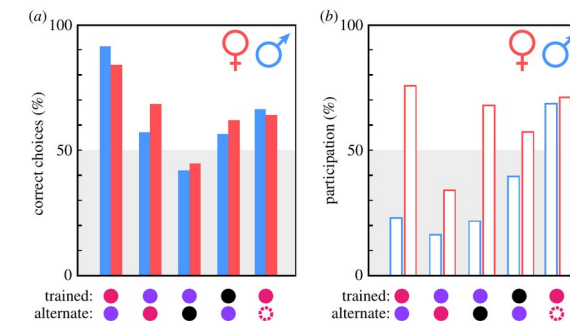
An **innate preference assay** measures an untrained choice rather than a response acquired through reward. Bok and colleagues offered *Haptosquilla trispinosa* two artificial burrows with dark or ultraviolet-illuminated entrances and measured which burrow the animal entered.

Dark burrows were selected over ultraviolet-illuminated burrows in roughly three-quarters or more of choices. When the alternatives were brightness-matched UVA and UVB burrows, the animals did not show a clear preference for one wavelength. Dark-versus-dark trials controlled for side preference.

Trained approach and untrained shelter selection therefore have different outcomes. Reduced participation, especially with UVB present, further separates willingness to respond from discrimination accuracy. The results support stimulus-dependent avoidance without identifying a neural aversion pathway or proving a particular natural communication function.



Bok et al. (2018), Fig. 1(g, h). Original innate burrow-choice apparatus and behavioral responses.



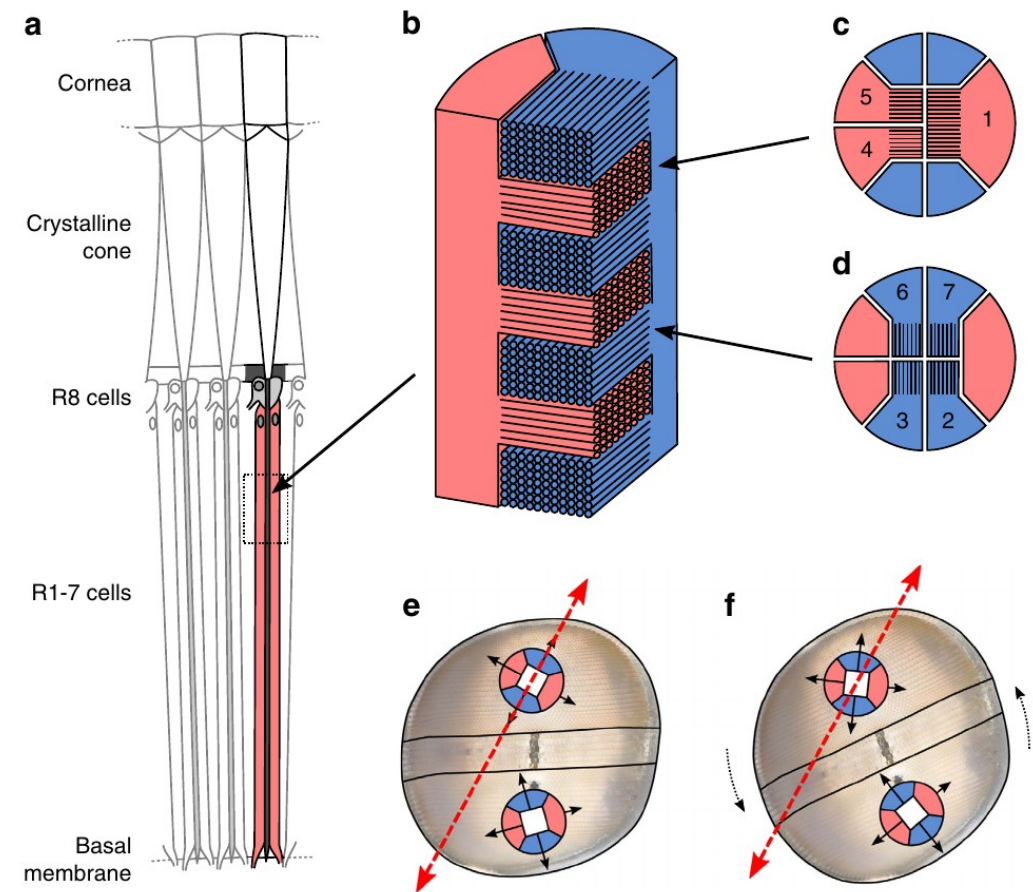
Bok et al. (2018), Fig. 2(a, b). Correct choices and participation differ across stimulus conditions.

Linear polarization adds information beyond wavelength

Polarization describes the organization of a light wave's electric-field oscillation. In **linearly polarized light**, the electric vector oscillates along one orientation. Two lights can share a wavelength spectrum and intensity while differing in polarization orientation.

Stomatopod photoreceptors contain ordered **microvilli**, tiny membrane projections that organize the light-sensitive pigment. Their response depends on the angle between the electric vector and the microvillar axes. Daly and colleagues related this geometry to measured eye movements in polarized stimulus fields.

Polarization sensing is therefore distinct from spectral discrimination. A visual response to orientation requires the appropriate receptor geometry and subsequent signal processing; it must not be inferred simply from the presence of many color-sensitive receptor classes.



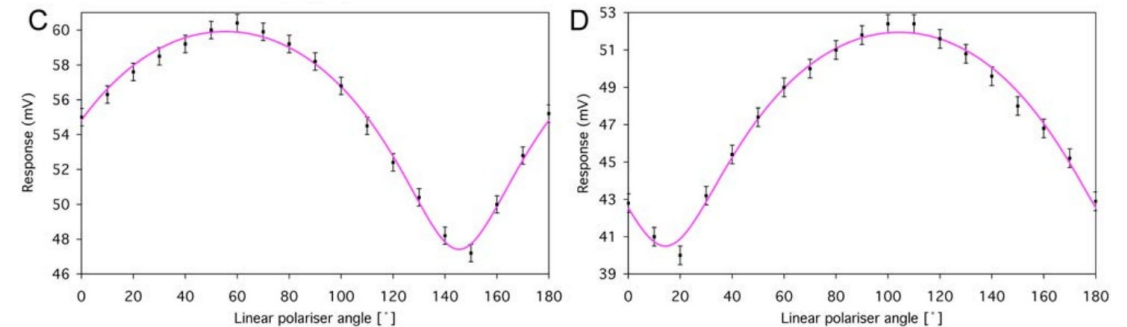
Daly et al. (2016), Fig. 2(a–f). The authors' published receptor geometry for linear-polarization sampling.

Microvillar orientation determines linear-polarization tuning

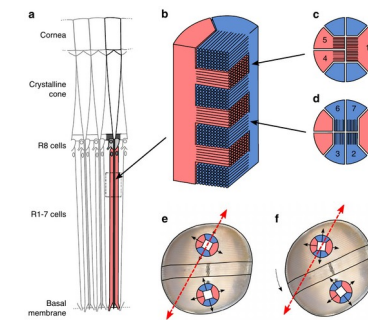
Polarization tuning is a response dependence on the electric-vector orientation of the stimulus. Kleinlogel and White recorded identified *Gonodactylus smithii* photoreceptors while rotating a linear polarizer in 10° steps. The recorded peaks followed the cells' microvillar orientation.

Representative dorsal and ventral R1 cells reached their largest responses at approximately 60° and 105° in the experiment's coordinates, a separation of 45° . Within an ommatidium, the complementary receptor group has an approximately orthogonal preferred axis.

The tuning curves establish directional receptor sensitivity under controlled stimulation. The plotted angles depend on eye and apparatus orientation; they are not universal horizontal and vertical preferences fixed to the world. Nonzero responses at unfavorable angles also limit any claim of perfect polarization selectivity.



Kleinlogel & White (2008), Fig. 4(C, D). Dorsal and ventral receptor voltage as the polarizer angle changes.



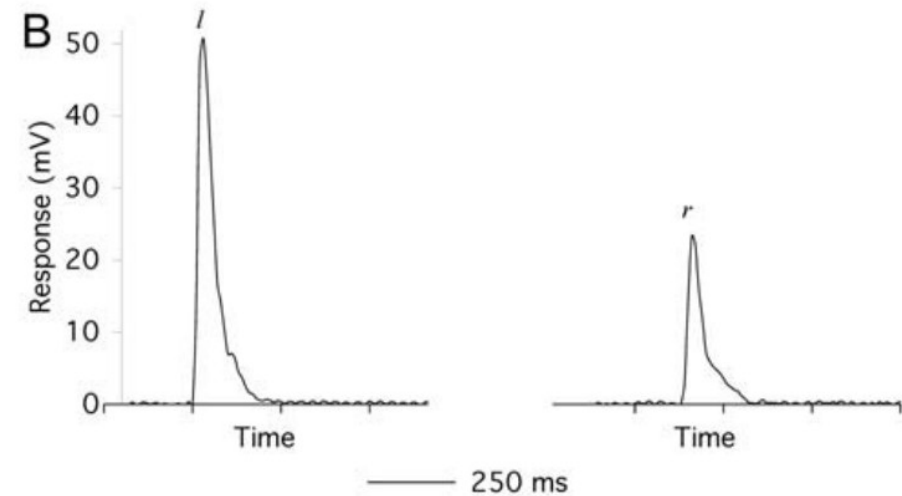
Daly et al. (2016), Fig. 2(a-f). Original geometry of alternating microvillar directions.

Brief flashes link polarization state to receptor voltage

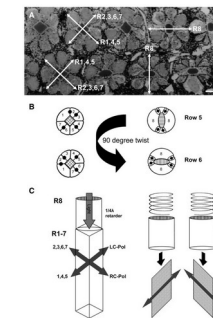
A **receptor potential** is a stimulus-dependent change in photoreceptor membrane voltage. Kleinlogel and White delivered 50-ms polarized flashes while a sharp electrode recorded *Gonodactylus smithii* cells. Linear and circular polarizing elements controlled the incident polarization.

A midband row-5 receptor produced different voltage responses to left- and right-circularly polarized flashes. Peak responses were used for intensity-response measurements. Histological identification after dye injection linked physiological responses to particular retinal cells.

The voltage difference establishes receptor-level sensitivity, not a behavioral choice or an independently isolated circular-polarization receptor molecule. The incident light is transformed by the overlying R8 structure before reaching the main rhabdom, so the measured response belongs to the combined optical and receptor system.



Kleinlogel & White (2008), Fig. 4(B). Different voltage responses to opposite circular states after 50-ms flashes.



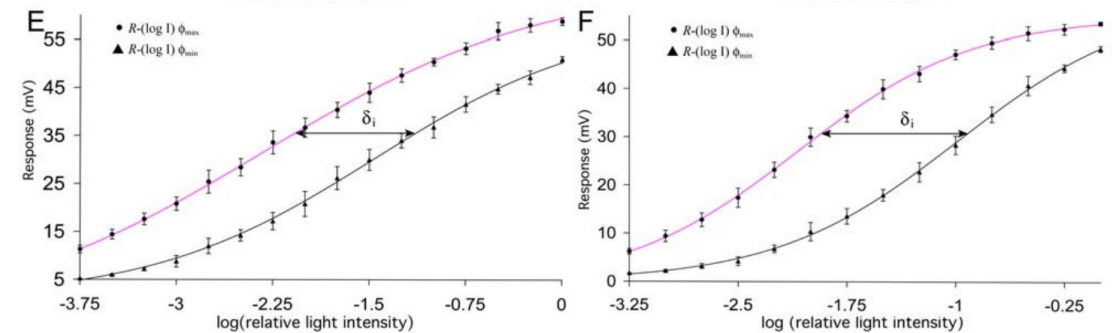
Chiou et al. (2008), Fig. 2(A-C). Original anatomy of the overlying R8 and R1-R7 system.

Intensity curves separate gain from polarization preference

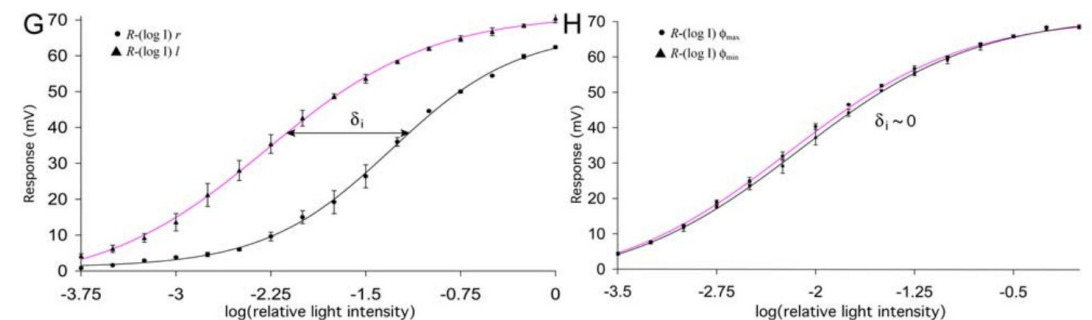
Polarization sensitivity can be estimated by comparing the light intensities needed for equal voltage responses under preferred and unfavorable polarization states. Kleinlogel and White measured intensity-response functions in *Gonodactylus smithii*, rather than using one response at a single intensity.

Hemispheric receptors required about an order-of-magnitude intensity difference between favorable and orthogonal linear orientations to produce comparable responses. Midband rows 5 and 6 similarly distinguished opposite circular states. Their linear-polarization intensity-response curves were nearly coincident under the tested conditions.

A shifted curve reflects differential sensitivity, whereas increasing voltage within either curve reflects intensity dependence. These measurements constrain the inputs available for neural comparison. They do not directly establish an opponent interneuron, its synapses or the animal's threshold for a behavioral polarization choice.



Kleinlogel & White (2008), Fig. 4(E, F). Intensity-response functions for preferred and orthogonal linear states.



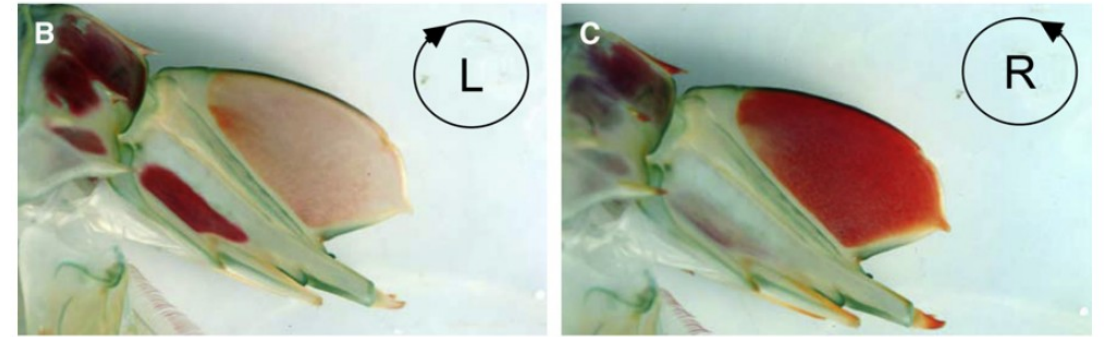
Kleinlogel & White (2008), Fig. 4(G, H). Circular-state preference and weak residual linear preference in midband cells.

Circular handedness can alter cuticular visual signals

In **circularly polarized light**, the electric-vector orientation rotates as the wave propagates. Opposite rotations define the two handedness states. Their distinction depends on a specified viewing convention; it is different from changing the angle of a linearly polarized stimulus.

Chiou and colleagues photographed male *Odontodactylus* cuticle through opposite circular-polarization analyzers. A telson region changed strongly in apparent contrast between the two analyzer states. The telson is the posterior body plate, often presented during stomatopod displays.

This optical measurement identifies a potential circular-polarization signal on the animal. It does not establish which social decision the signal controls. The proposed private communication role remains an ecological interpretation, whereas the state-dependent reflected contrast is measured evidence.



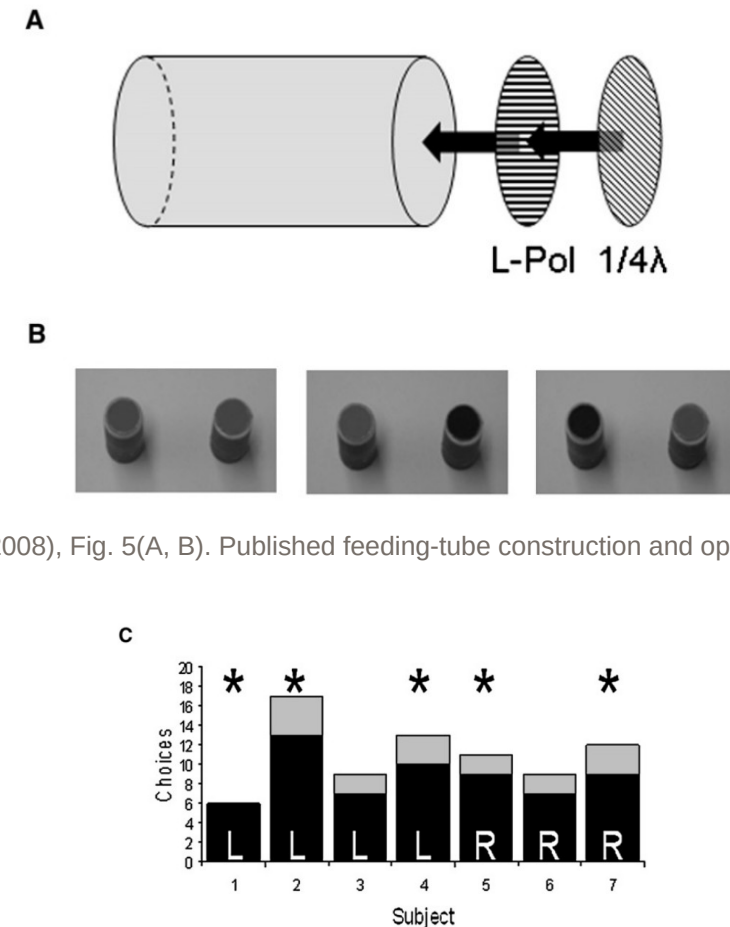
Chiou et al. (2008), Fig. 1(B, C). The same male telson region viewed through opposite circular analyzers.

Rewarded choices establish circular-state discrimination

Operant conditioning associates an animal's action with a consequence. Chiou and colleagues trained *Odontodactylus scyllarus* to select feeding tubes reflecting either left- or right-circularly polarized light. Food was placed at the open end during training.

Test tubes were presented without food, with the reflecting end facing the animal. Their construction combined a linear polarizer, a quarter-wave retarder and a reflective backing. Trained animals preferentially selected the circular state associated with reward, rather than choosing only by the unfiltered appearance of the tubes.

The choices establish behavioral discrimination of circular handedness in this task. They go beyond receptor sensitivity while remaining distinct from a demonstration of natural mate recognition, signaling function or the complete neural pathway producing the choice.



Chiou et al. (2008), Fig. 5(A, B). Published feeding-tube construction and opposite-analyzer photographs.

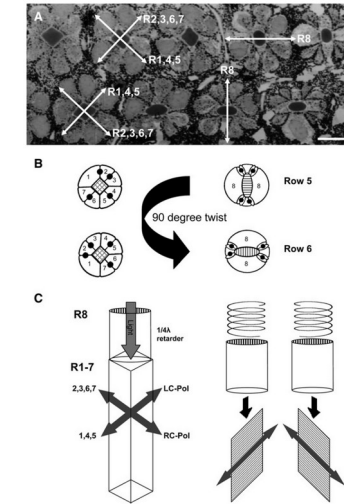
Chiou et al. (2008), Fig. 5(C). Individual tube choices after training to circular handedness.

R8 transforms circular states into orthogonal linear inputs

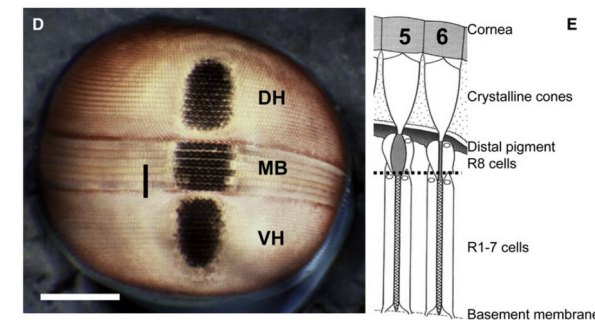
A **quarter-wave retarder** introduces a relative phase delay of one-quarter wavelength, or 90° , between orthogonal electric-field components. In *Odontodactylus* midband rows 5 and 6, an ordered R8 rhabdomere lies above two orthogonal R1–R7 receptor groups.

R8 microvilli are oriented about 45° relative to the underlying absorption axes. Circular states of opposite handedness are converted into linear states at mutually orthogonal angles. The underlying receptor groups can then respond differently through their ordinary directional absorption.

The mechanism links tissue geometry to receptor voltage without requiring a circular-selective pigment. Its optical effectiveness depends on retardance and wavelength. Similar-looking R8 structures in other stomatopods must not be assumed to provide identical conversion without species-specific optical or physiological measurements.



Chiou et al. (2008), Fig. 2(A–C). Unchanged published anatomy and quarter-wave conversion mechanism.



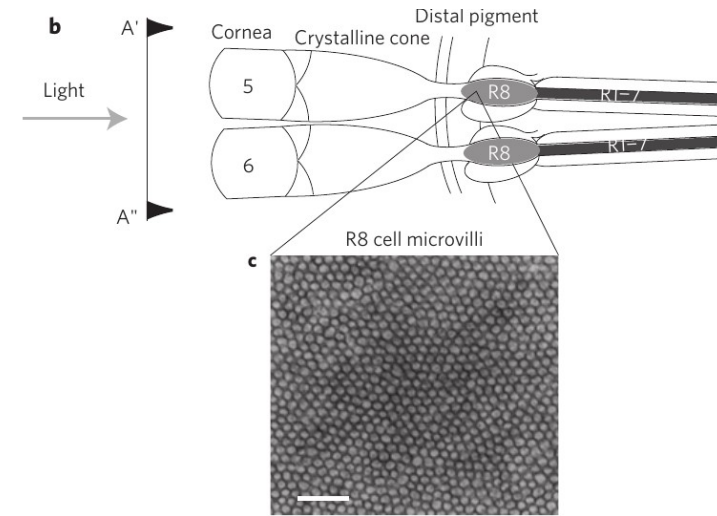
Chiou et al. (2008), Fig. 1(D, E). Location of polarization rows and their optical tiers.

Ordered microvilli produce optical birefringence

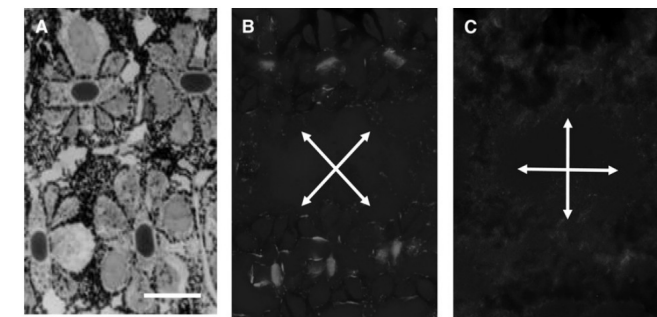
Birefringence is a difference in refractive behavior for orthogonal polarization components. A **refractive index** describes how light propagates through a material relative to vacuum. Roberts and colleagues examined the ordered R8 membrane structure of *Odontodactylus scyllarus*.

Electron microscopy gave microvillar inner and outer radii of approximately 26 and 40 nm. These dimensions are much smaller than visible wavelengths. Polarizing-microscope observations from Chiou and colleagues also changed when the analyzer axes rotated relative to the tissue.

An orientation-dependent bright region between crossed polarizers supports birefringence rather than simple depolarization. It does not by itself prove a quarter-wave phase delay. Quantitative transmission measurements and receptor recordings are needed to connect the microscopic architecture with circular-state discrimination.



Roberts et al. (2009), Fig. 1(b, c). R8 tissue position and electron micrograph of its microvilli.



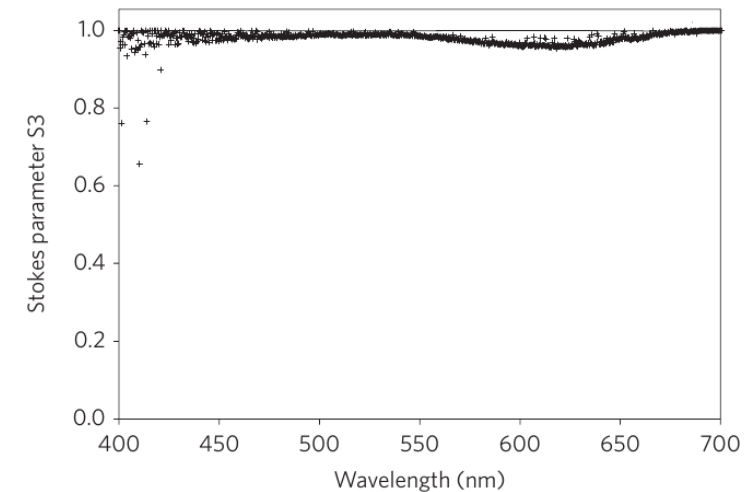
Chiou et al. (2008), Fig. 3(A–C). Tissue and crossed-polarizer views at different orientations.

R8 conversion spans the visible spectrum

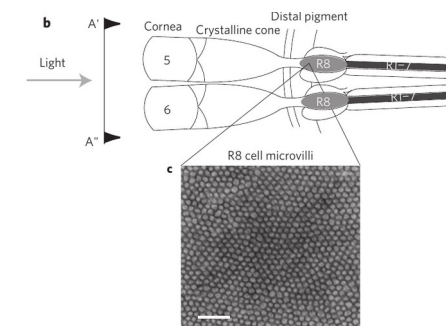
Achromaticity means that an optical operation changes relatively little with wavelength. Roberts and colleagues measured transmission through 10- μm cryosections of *Odontodactylus scyllarus* R8 rhabdomeres and scaled the optical measurements to the approximately 150- μm receptor length.

Light entered with linear polarization at 45° to the microvillar axis. The transmitted circular component, described by the **Stokes parameter S3**, remained close to that of strongly circular light across much of the 400–700 nm range. A fitted optical model reproduced this wavelength dependence.

Broadband conversion helps preserve opposite circular-state signals across the broad absorption spectrum of the underlying receptors. The curve describes optical transmission, not a receptor firing rate or measured behavioral threshold at each wavelength.



Roberts et al. (2009), Fig. 2. Measured circular component after transmission and the fitted optical model.



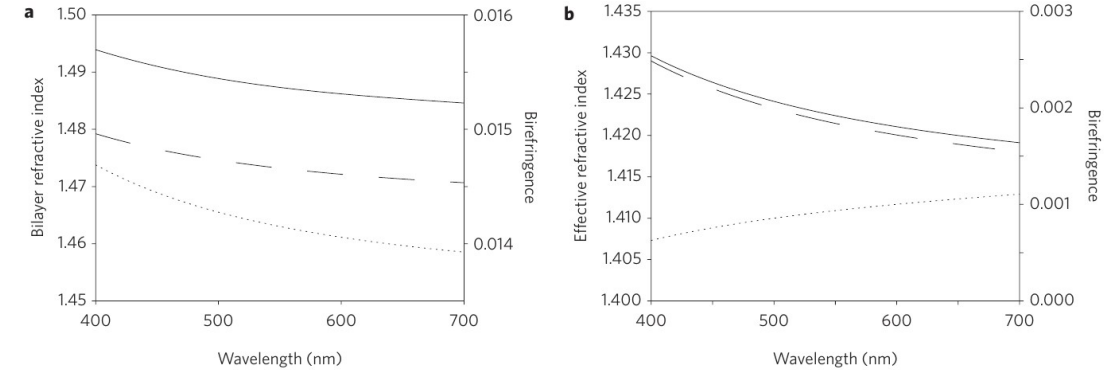
Roberts et al. (2009), Fig. 1(b, c). The R8 tissue whose transmission was measured.

Membranes and geometry jointly shape retardance

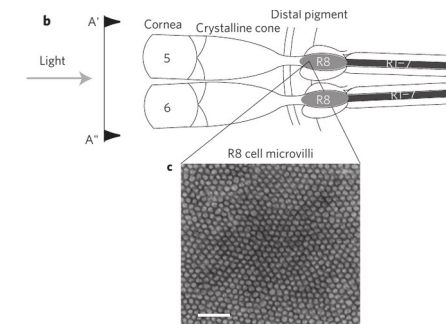
Intrinsic birefringence arises from the material's directional optical properties. **Form birefringence** arises from the arrangement of different materials at small spatial scales. Roberts and colleagues modeled both contributions in the membranes and cytoplasm of the R8 rhabdomere.

The model fixed the measured microvillar dimensions and receptor length while fitting membrane refractive properties. Their interaction produced the wavelength-dependent effective birefringence needed for nearly constant quarter-wave retardance. The fitted material properties agreed with plausible membrane refractive values.

Agreement between model and transmission supports this biophysical account of the optic. It does not identify a dedicated neurotransmitter, ion channel or central comparator: the modeled operation transforms light before the downstream electrical signal is generated.



Roberts et al. (2009), Fig. 3(a, b). Published intrinsic and effective refractive-index model functions.



Roberts et al. (2009), Fig. 1(b, c). The measured microvillar geometry underlying the model.

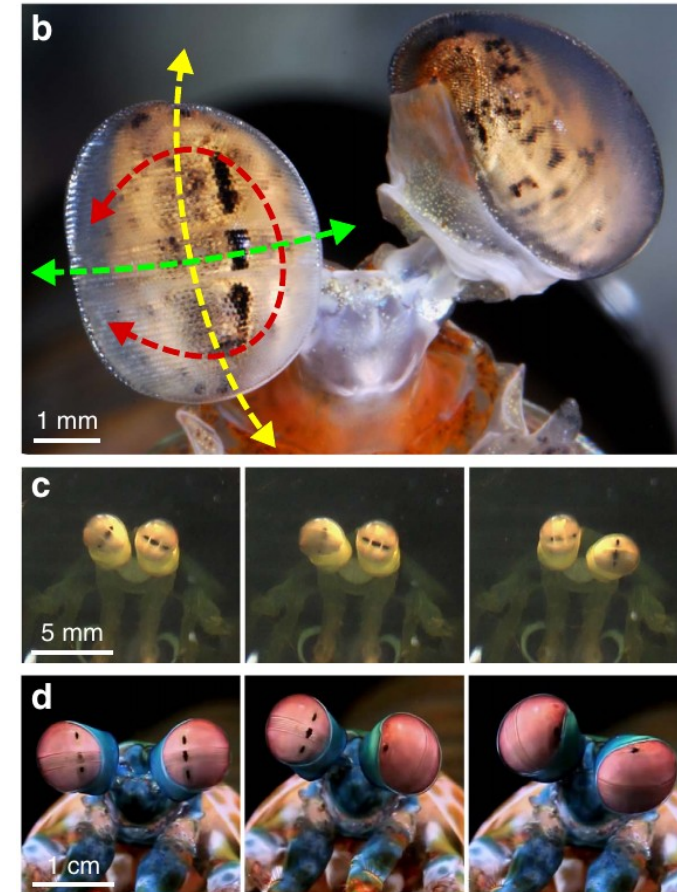
Torsion rotates the eye's polarization-sampling axes

Torsion, or roll, rotates the eye about its viewing axis.

Stomatopods can also move their eyes in pitch and yaw, changing the direction of gaze. Daly and colleagues tracked three-dimensional eye orientation in *Gonodactylus smithii* and *Odontodactylus scyllarus*.

Large torsional movements can reach approximately 90° in these species. Rolling the eye changes the orientation of its microvillar sampling axes relative to a polarized scene, even when the animal continues viewing the same object. The two eyes can move partly independently.

The movement geometry makes active polarization sampling possible. Spontaneous rotation alone does not prove that the animal maximizes a particular visual signal; controlled stimuli and measurements of the resulting receptor alignment are required to test that function.



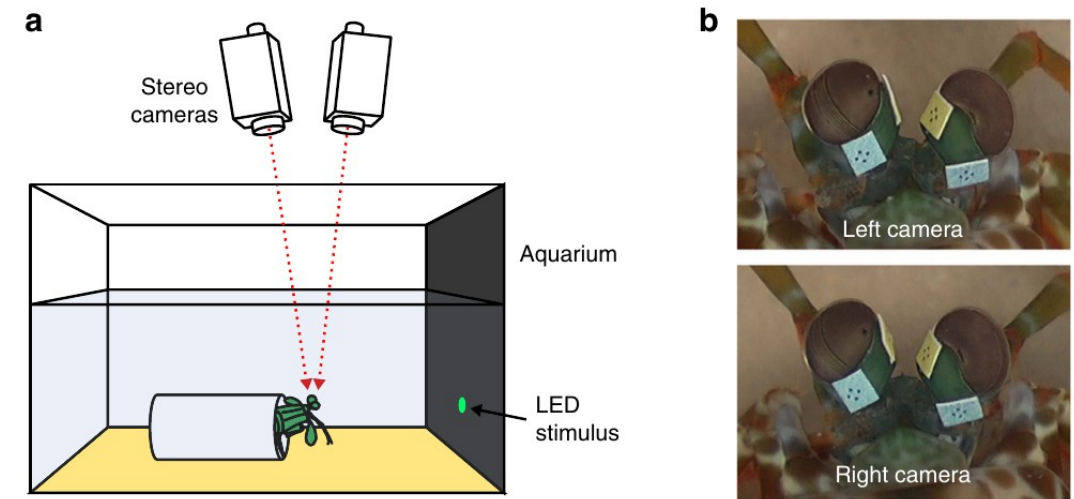
Daly et al. (2016), Fig. 1(b–d). Published eye rotation axes and successive torsional positions.

A polarized LED elicits torsional eye rotations

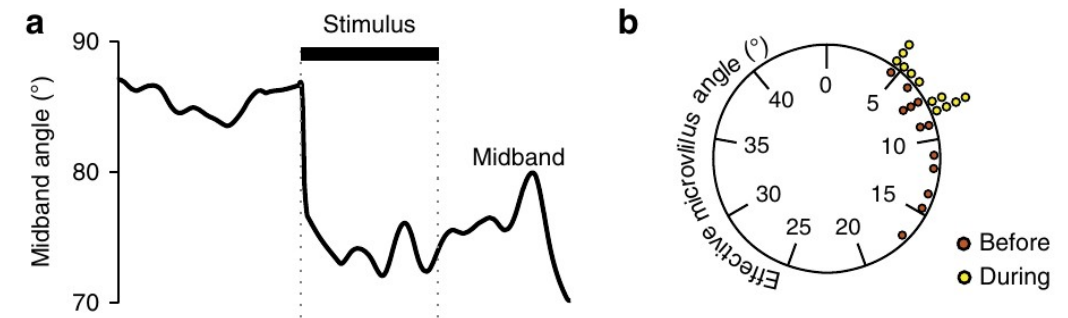
Daly and colleagues presented *Gonodactylus smithii* with a green LED peaking at 501 nm through a 4-mm aperture against a dark background. Its light was strongly linearly polarized, with either vertical or diagonal electric-vector orientation. Two cameras measured eye rotation.

The 3-s presentation commonly elicited a torsional response. Initial rotations averaged about 15° , and the responding eye moved its nearest hemispheric receptor axis toward the stimulus polarization angle. Stimulus timing and three-dimensional eye position were recorded together.

This experiment links a controlled polarized cue to measured motor output. It does not directly record the oculomotor command or identify the neurons causing the rotation. The resulting receptor alignment is derived from anatomical geometry and measured eye position.



Daly et al. (2016), Fig. 5(a, b). Original LED presentation and stereo-camera recording arrangement.



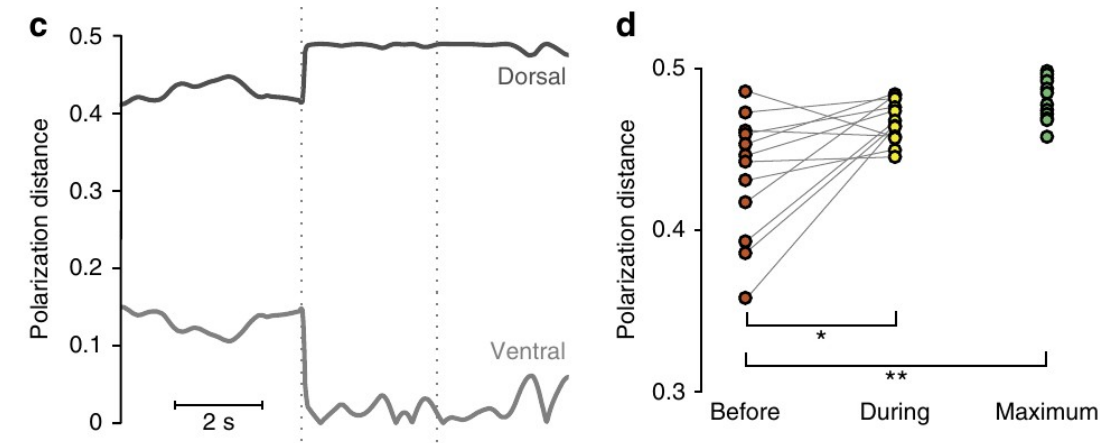
Daly et al. (2016), Fig. 3(a, b). Measured eye rotation and receptor-axis alignment during the stimulus.

Eye positions predict increased polarization contrast

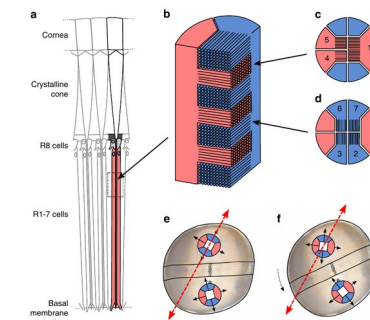
Polarization distance is a model-based measure of contrast between two polarization states available to a specified receptor system. Daly and colleagues calculated it from measured eye orientation, published receptor geometry and the polarization of the LED and its background.

During LED presentation, predicted hemispheric polarization contrast increased toward the model's favorable alignment. The dorsal and ventral receptor populations differed in their calculated contrast because their axes were offset by 45° . The movement did not similarly maximize the modeled midband polarization signal.

These results support active enhancement of a hemispheric signal. The contrast values are calculations constrained by measurements, rather than recorded interneuron voltage or proof that the animal explicitly computes the same quantity.



Daly et al. (2016), Fig. 3(c, d). Published model-based contrast estimates before and during the LED.



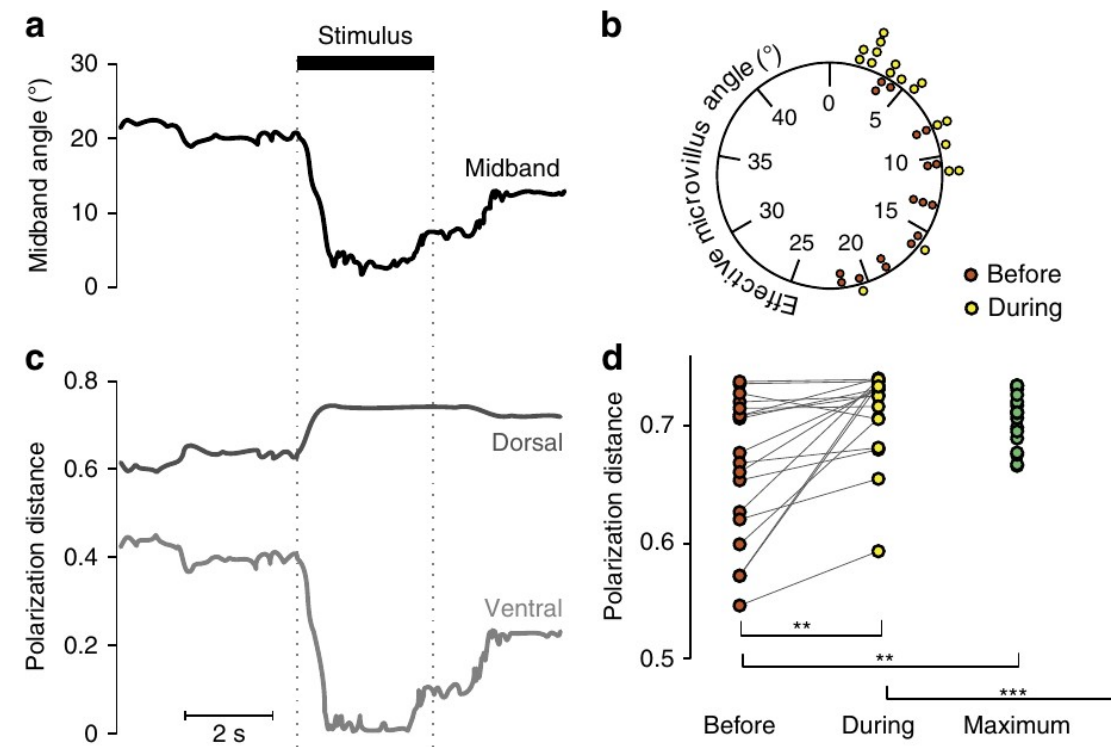
Daly et al. (2016), Fig. 2(a-f). Anatomical axes used in the contrast calculation.

Polarized looming recruits visual sampling

A **looming stimulus** expands in angular size, mimicking an approaching object. Daly and colleagues modified an LCD display to present a growing disk whose polarization differed from its background while color and intensity remained uniform. Illumination peaked at 501 nm.

Odontodactylus scyllarus received disks expanding to a visual angle of 10° , with an object–background polarization-angle difference of 61.2° . Torsional responses changed the modeled hemispheric contrast. Tests at several display tilts checked whether the effect depended on one screen geometry.

The result supports a motor response to a polarization-defined object under controlled conditions. It does not demonstrate an unrestricted predator-avoidance trajectory, and the reconstructed contrast still depends on the receptor model rather than a simultaneously recorded central signal.



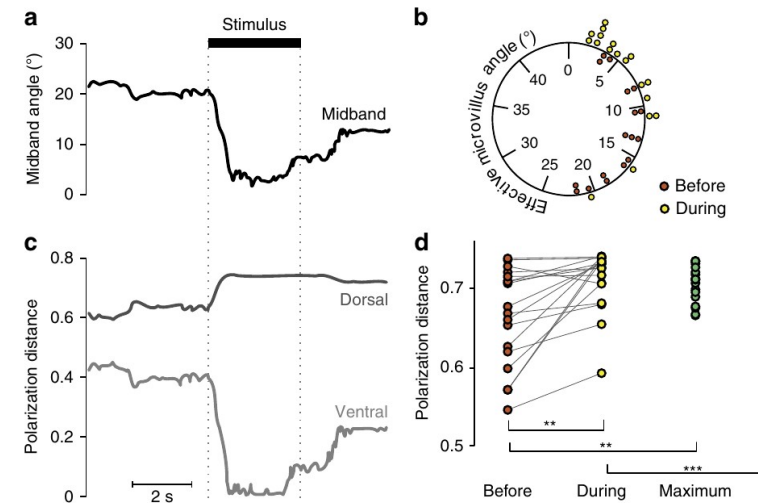
Daly et al. (2016), Fig. 4(a–d). Original eye-position records and contrast estimates during polarized looming.

Maximum contrast need not align with the object alone

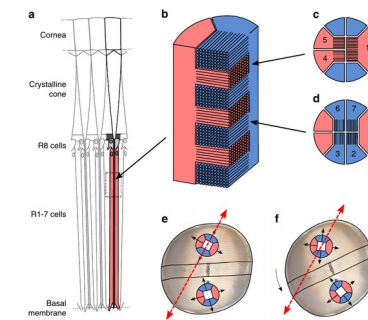
The best receptor orientation depends on both object and background polarization. For an object against an unpolarized background, aligning one axis with the object can maximize contrast. With a differently polarized background, that orientation need not produce the largest difference.

Daly and colleagues compared measured *Odontodactylus scyllarus* rotations during looming with two predictions: maximal object–background contrast and alignment with the object's polarization alone. Eye positions were more consistent with the contrast-maximizing account.

Torsion therefore supports context-dependent active sampling rather than a fixed rule to point one axis at every polarized object. The comparison tests optical consequences of movement; the neuronal representation of the background and the motor computation selecting an orientation remain unresolved.



Daly et al. (2016), Fig. 4(a–d). Measured positions favor contrast over the object-only alignment prediction.



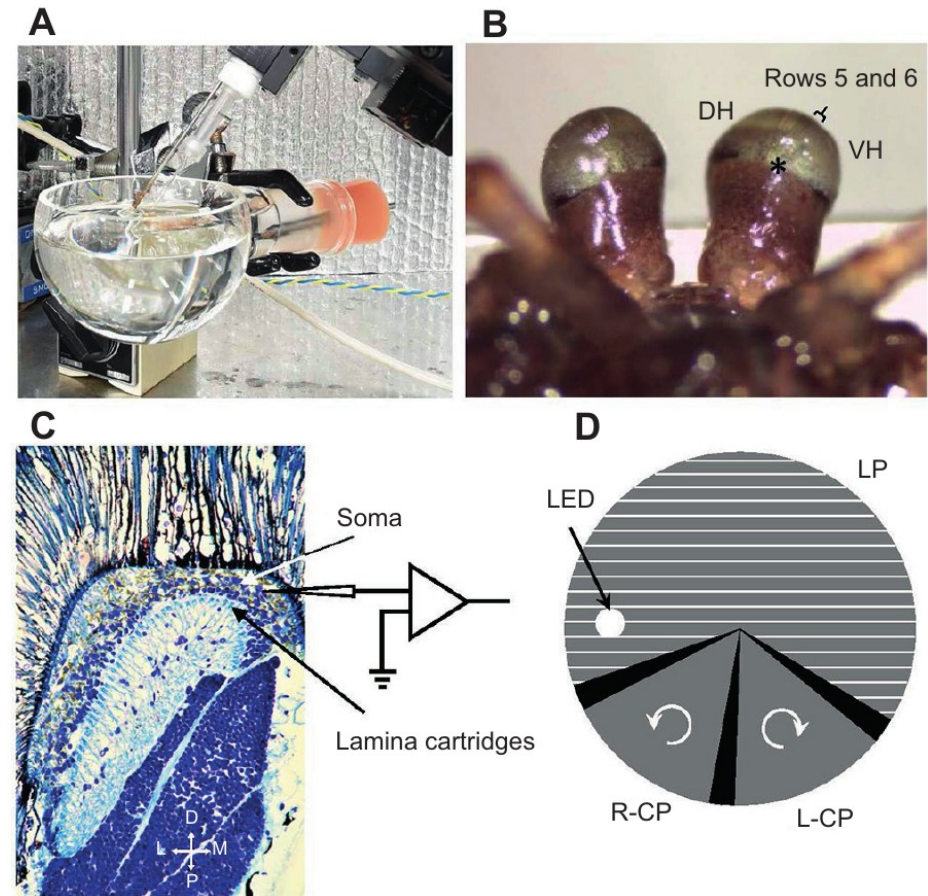
Daly et al. (2016), Fig. 2(a–f). Published geometry of hemispheric sampling axes.

Sharp electrodes record retinal and lamina activity

Chiou and Wang recorded intracellular membrane potentials from the retina and lamina of *Haptosquilla pulchella*. A chilled animal was secured in oxygenated seawater, and a small eyestalk opening allowed a sharp electrode to approach the tissue near the midband.

Electrode resistances were 80–120 M Ω , and voltage was sampled at 10 kHz. A rotating polarizing assembly presented linear orientations followed by opposite circular states in a sequence lasting about 1.75 s. Reversing rotation provided a check on stimulus-order effects.

The preparation links specified polarization states to activity beyond the receptor layer. Cell categories were assigned partly from recording location and response profile, so physiological labels must not be treated as fully traced anatomical identities or identified transmitter classes.



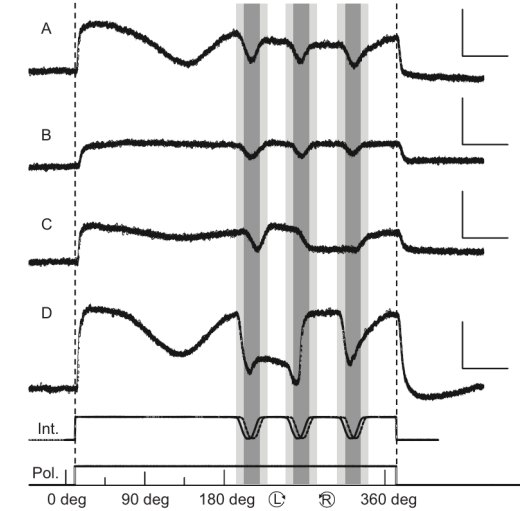
Chiou & Wang (2020), Fig. 1(A–D). Original recording chamber, eyestalk access and lamina section.

Receptors can combine linear and circular tuning

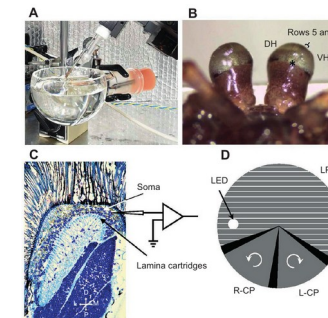
Elliptically polarized light includes both linear and circular components; purely linear and circular states are limiting cases. Chiou and Wang recorded *Haptosquilla pulchella* receptors during the same sequence of changing linear orientation and opposite circular handedness.

Some cells primarily tracked linear angle, others changed little with polarization, and another group preferred one circular handedness while retaining angle-dependent fluctuations during linear stimulation. The published traces carry 10-mV vertical and 250-ms horizontal scale bars.

Combined sensitivity supports receptors tuned to elliptical rather than perfectly circular states. It prevents transferring ideal circular selectivity from *Odontodactylus* directly to *Haptosquilla*. The inferred row assignments were based on function and access geometry, leaving the exact identified-cell correspondence less secure than a dye-filled recording.



Chiou & Wang (2020), Fig. 2(A–D). Original receptor traces for changing linear angle and circular handedness.



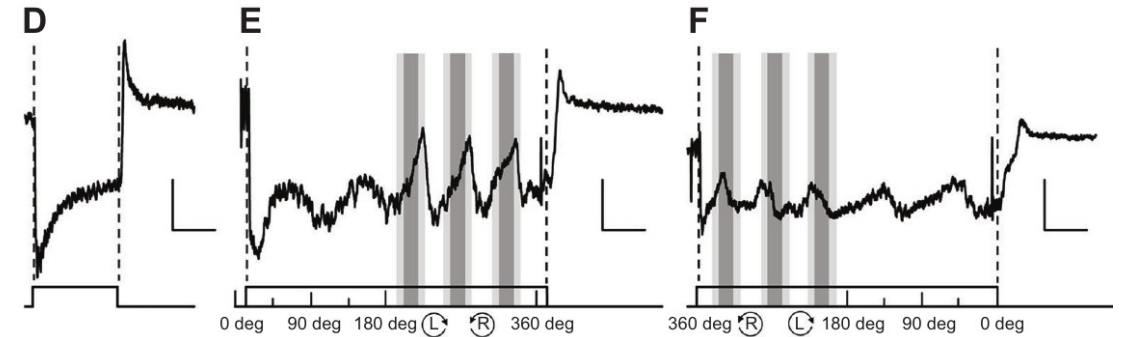
Chiou & Wang (2020), Fig. 1(A–D). Retinal access and the sampled lamina region.

Lamina responses differ from single-receptor tuning

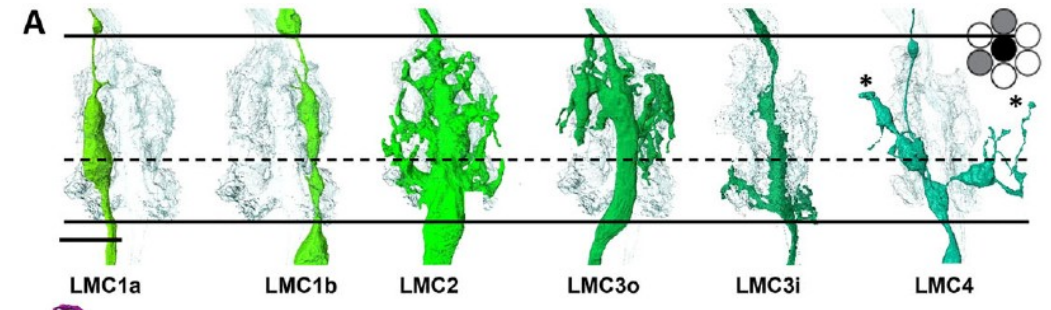
A **lamina monopolar cell** is an interneuron with a main axon traversing the lamina and branches within its processing layers. Chiou and Wang identified monopolar-cell-like responses in *Haptosquilla pulchella* from recording depth and characteristic graded voltage changes.

A 0.5-s white-light pulse produced hyperpolarization, a voltage shift toward a more negative membrane potential. Polarization-sensitive examples had two response episodes as the linear electric vector swept through 180° , including approximately 45° and 135° in one recording. Reverse rotation altered waveform shape while preserving its period.

The responses support convergence from differently oriented inputs. They do not establish every synaptic sign or prove that these physiological cells are a particular reconstructed monopolar subtype. Their voltage pattern differs from the single preferred-axis response of a photoreceptor.



Chiou & Wang (2020), Fig. 3(D–F). Monopolar-cell-like voltage responses to flashes and reversed polarization sequences.



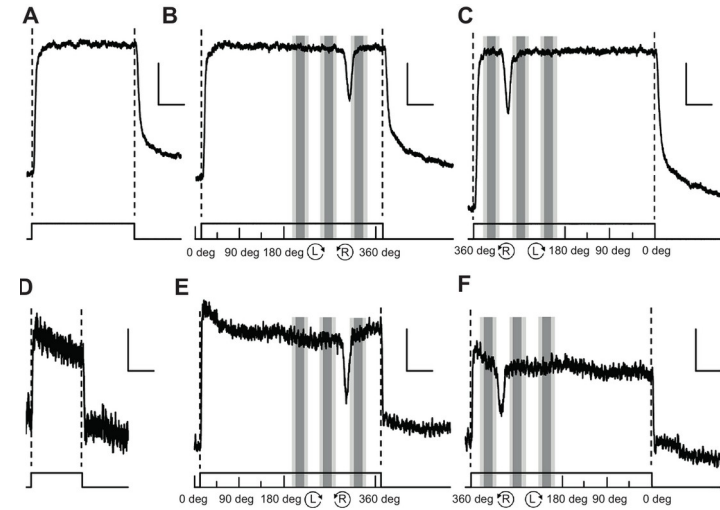
Streets et al. (2025), Fig. 5(A). Published monopolar-cell and neighboring interneuron reconstructions.

Lamina cells can favor circular handedness selectively

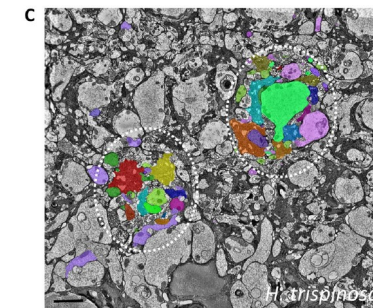
Chiou and Wang found lamina-region cells in *Haptosquilla pulchella* whose responses differed specifically during right-circularly polarized stimulation. A brief white-light pulse depolarized these cells, increasing their membrane potential relative to rest.

During the rotating stimulus sequence, right-circular light elicited a transient decrease from the depolarized level. Changing the stimulus to an off-axis position increased noise but preserved the qualitative response pattern. The brief intensity reductions at filter boundaries did not produce the same characteristic response.

Selectivity beyond the retina supports separation of linear and circular signal components in early visual processing. The cells were not fully morphologically identified, and the recording does not reveal a complete opponent circuit, its transmitter or its contribution to the learned feeding-tube choice.



Chiou & Wang (2020), Fig. 4(A–F). Original circular-sensitive responses under on-axis and off-axis stimulation.



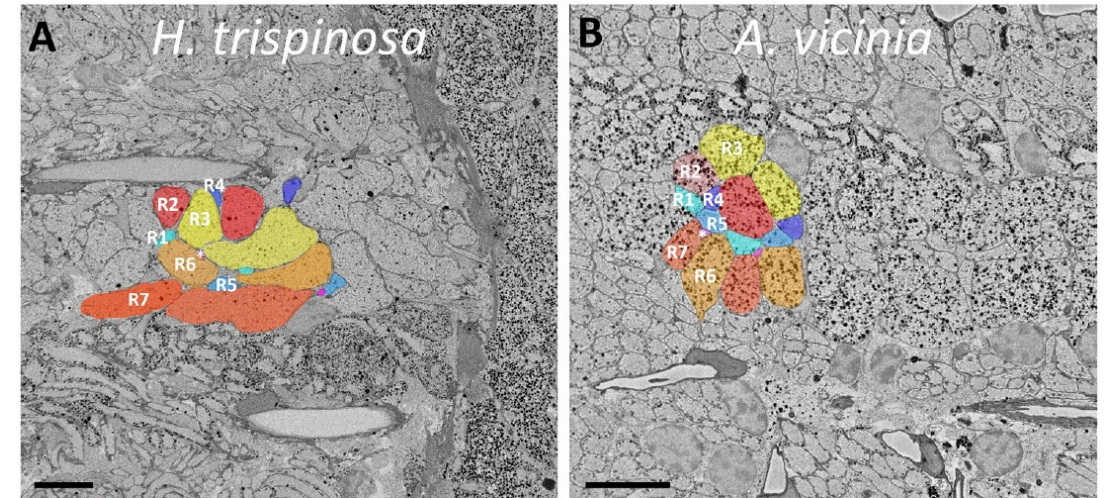
Streets et al. (2025), Fig. 5(C). Published lamina micrographs with identified neuronal profiles.

Volume microscopy reconstructs lamina input columns

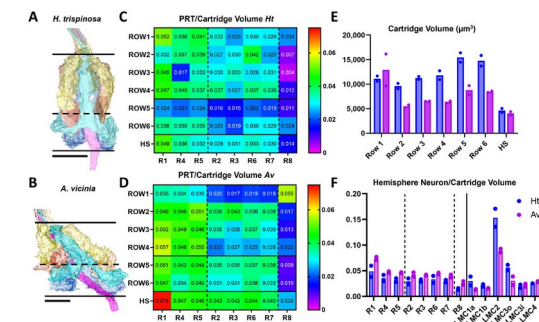
Serial block-face electron microscopy repeatedly images a tissue block as thin layers are removed, producing a three-dimensional volume. Streets and colleagues used it to follow photoreceptor axons and lamina interneurons in *Haptosquilla trispinosa* and *Alachosquilla vicina*.

High-resolution stacks used 50-nm section steps, with image pixels approximately 10–15 nm across. Identified profiles were traced through successive images to reconstruct cartridge organization. Distal retinal cell identities constrained the assignment of the axons entering the lamina.

The volumes establish spatial relationships and branching patterns at finer resolution than conventional tissue staining. They were insufficient to identify all synapses in these preparations, so reconstructed proximity remains distinct from a directly measured connection, response sign or synaptic strength.



Streets et al. (2025), Fig. 3(A, B). Original electron micrographs with author-labeled receptor profiles.



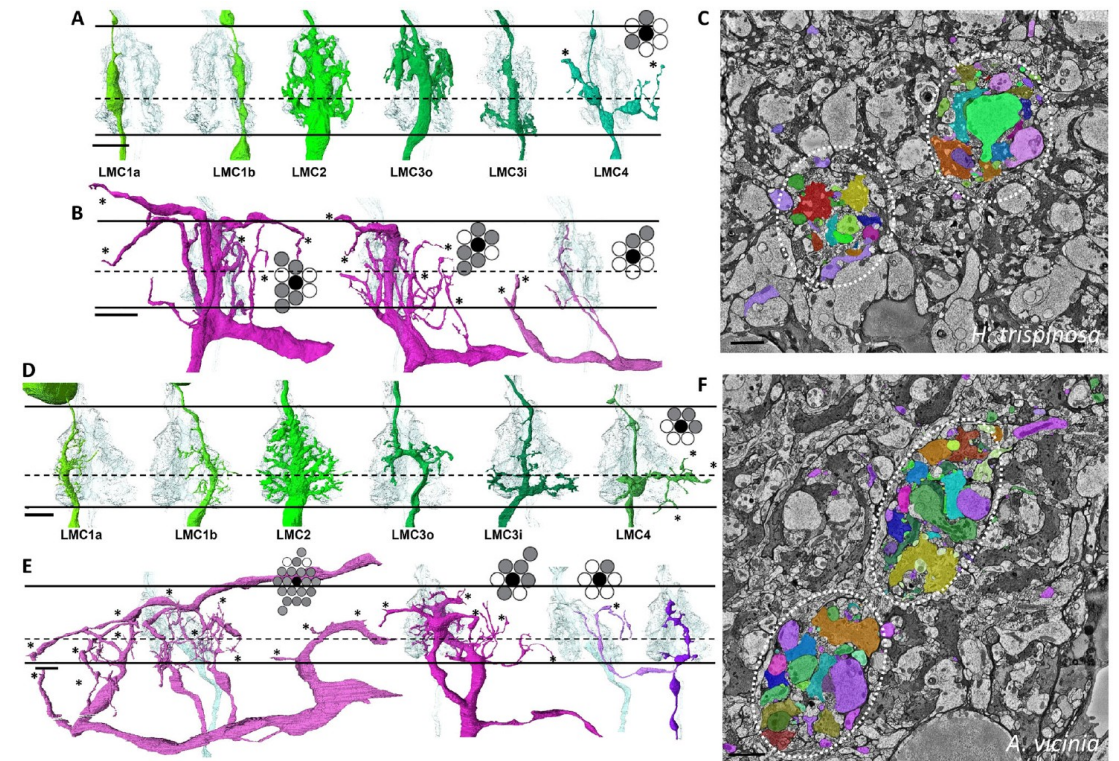
Streets et al. (2025), Fig. 4(A–F). Author-reconstructed hemisphere photoreceptor terminals.

Monopolar branching preserves distinct candidate inputs

Streets and colleagues reconstructed multiple monopolar-cell types in individual lamina cartridges. LMC3 outer and inner cells concentrated branches in different external plexiform layers, while LMC2 extended branches through both layers. LMC1 cells occupied complementary sides of a cartridge.

These patterns occurred in two stomatopod species with different visual ecologies. Layer-restricted cells are candidates for retaining distinct spectral or polarization inputs, while cells spanning layers could integrate a broader set of receptor signals. The published reconstructions include 10- μ m scale bars.

Opponency requires more than separated inputs: one stimulus must affect the output differently from another through a defined interaction. Morphology supports candidate circuitry but does not establish inhibitory synapses, the transmitter involved or the precise correspondence with physiologically classified cells.



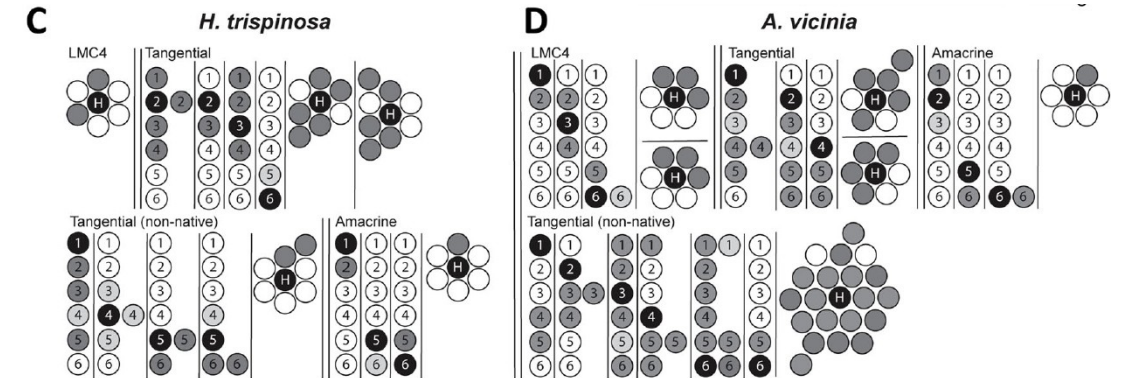
Streets et al. (2025), Fig. 5(A–F). Comparative *Alachosquilla* interneuron reconstructions.

Cross-cartridge branches can integrate channels

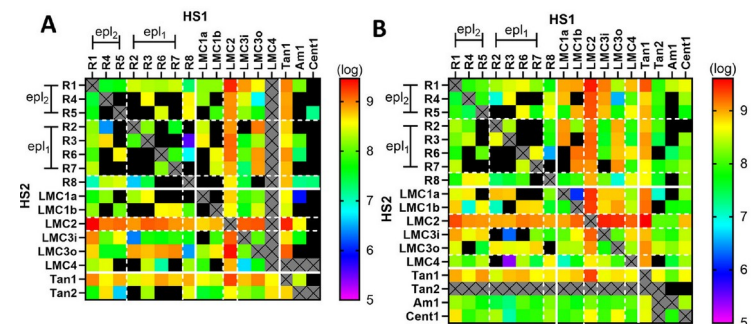
Tangential cells have branches extending across processing columns. Streets and colleagues traced such cells and other interneurons beyond their parent lamina cartridges, finding routes between nearby midband rows as well as within rows and hemispheric regions.

To estimate possible contacts, the authors measured reconstructed volume overlap after approximately 50-nm dilation of segmented surfaces. Their matrices summarize proximity between identified processes. Midband branches often extended within the same visual column, potentially linking different spectral sampling rows.

These are candidate integration routes. Geometric overlap is a necessary spatial opportunity for many synapses but is not proof of a synapse, its direction or its strength. The heat-map values therefore must not be relabeled as measured synaptic weights or confirmed physiological connectivity.



Streets et al. (2025), Fig. 6(C, D). The authors' maps of branches extending into neighboring cartridges.



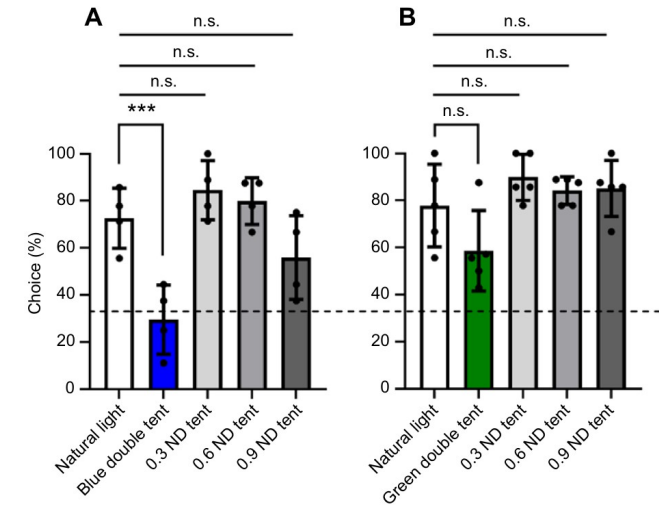
Streets et al. (2025), Fig. 6(A, B). Original proximity-derived overlap matrices, not verified synaptic connections.

The complete opponent circuit remains unresolved

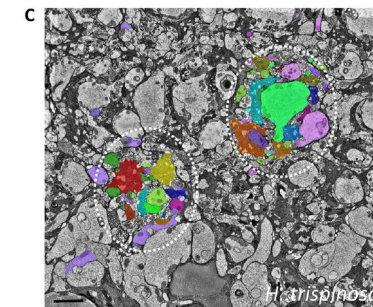
Wang and Marshall varied illumination intensity as well as spectral composition in *Haptosquilla trispinosa*. Double-layered blue tents reduced blue-target discrimination, whereas neutral-density dimming preserved useful blue and green choices. Green choices exceeded 80% under the dimmest tested neutral illumination.

Together with restricted-spectrum discrimination, these controls support spectral comparison beyond a simple brightness rule. Streets and colleagues provide anatomical candidate pathways, while Chiou and Wang recorded transformed polarization responses in a different *Haptosquilla* species. The studies test complementary but distinct levels of processing.

A hybrid account of spectral recognition and opponency remains supported as a proposal, not a completed circuit. The comparing neurons, transmitter identities, postsynaptic receptors, ion channels and causal links to particular color decisions have not been established by these experiments.



Wang & Marshall (2025), Fig. 8(A, B). Color-versus-grey choices under dim colored and neutral illumination.



Streets et al. (2025), Fig. 5(C). Original lamina tissue profiles that constrain candidate pathways.

Key takeaways

Many inputs, coarse discrimination Narrow spectral receptors do not guarantee fine color resolution; Haptosquilla wavelength thresholds were approximately 12–25 nm in the tested choice assay.

Polarization-sensitive microvilli Aligned membranes produce directional absorption, and distinct hemispheric receptor axes provide complementary linear-polarization inputs.

Active polarization sampling Torsional eye movements change receptor alignment; object–background polarization contrast can favor an orientation different from object alignment alone.

Pigments plus filters Tiered receptors and serial filters shape effective spectral sensitivity; two UV pigments can support several receptor classes through cone filtering.

R8 optical conversion Quarter-wave retardance transforms opposite circular states into orthogonal linear inputs for the underlying receptors in species with the required geometry.

Processing and its limits Behavior supports spectral comparison and lamina recordings support polarization transformations; hybrid color decoding and the complete synaptic mechanism remain unresolved.